ANALYSIS

The data economy of public proteomics over thirteen years of PRIDE

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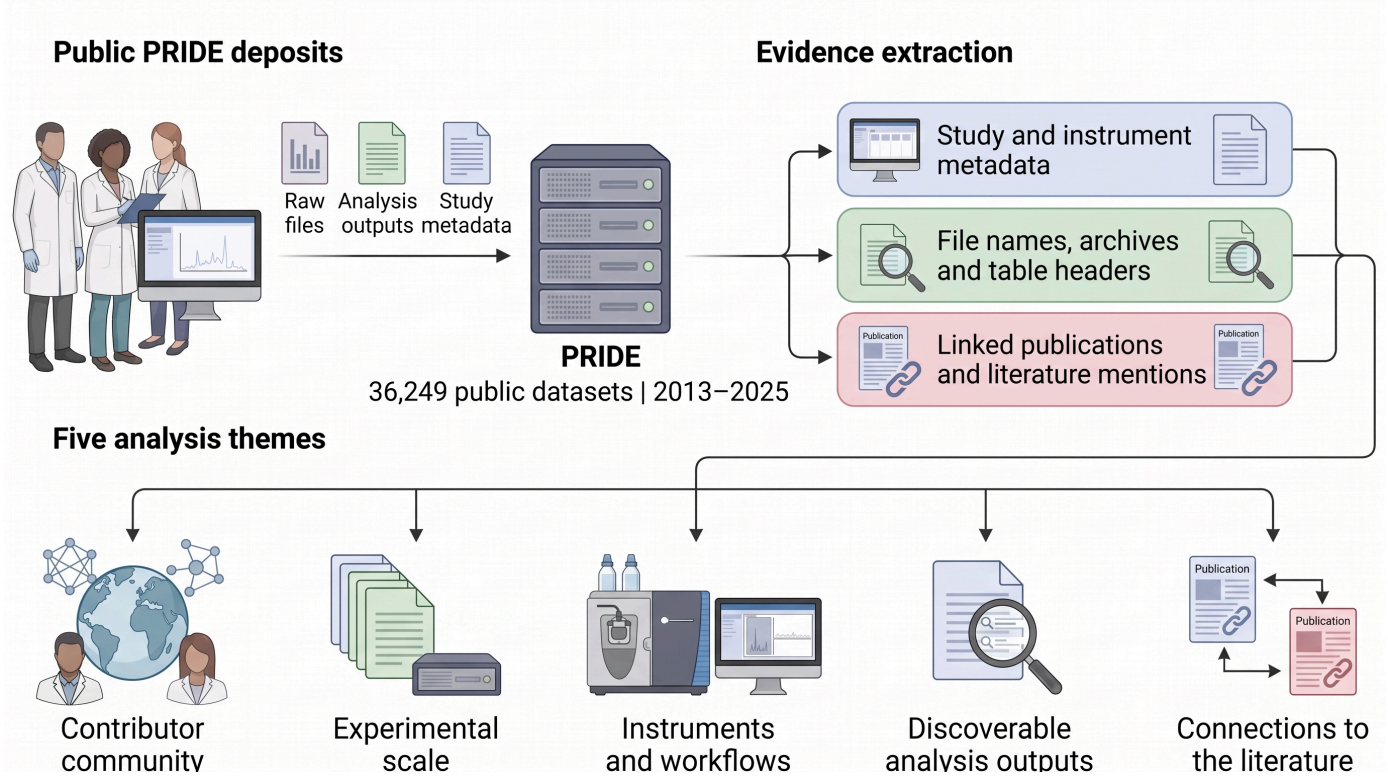
Pre-submission review · October 2026

PRIDE's public archive records how proteomics research changes as its community grows. We examined all 36,249 datasets released between 2013 and 2025 using study records, deposited files and links to published papers. The contributor community broadened, and linked papers appeared in a wider range of journals. Large datasets became more numerous, although the median number of mass-spectrometry runs per dataset fell. New instruments and software appeared in deposits alongside older ones. We recognised peptide or protein identification results in 80% of 2025 datasets by reading files produced by individual analysis programs; 16% contained results in community-standard formats. We detected no association between those standard formats and dataset mentions in the other papers we analysed. Mentions alone cannot show that the data were analysed again. Searching the formats used by individual analysis programs could help readers find more of the results already shared through PRIDE.



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Introduction

PRIDE makes the accumulated work of the proteomics community available for questions beyond those that motivated each experiment. As the largest repository in the ProteomeXchange network, it preserves mass-spectrometry proteomics data, provides public access to deposited files and connects them to the published literature[1,2]. This infrastructure supports what we call the data economy of public proteomics: the sharing, discovery and subsequent use of experimental data, considered here through their scientific circulation rather than their monetary value.

The files preserved in PRIDE also provide a record of how proteomics research has changed as its community has grown. Raw files and instrument labels describe the experiments, while logs and parameter files identify the software used to analyse them. Read alongside release dates and publication links, these records connect the growth of the archive to the scale of experiments, the workflows researchers use and the results they make available. They offer a way to study research practice across the community through evidence retained in its public deposits.

Previous studies have examined the impact of public proteomics data through citations[3] and assessed individual deposits through reanalysis of six projects[4] and an audit of 500 datasets[5]. An archive-wide analysis can extend this work by placing individual deposits within the changing practices of the contributing community. It can also connect developments such as large cohort studies[6,7] and community-standard result formats[8] to the experiments and files represented in the public record.

Here, we examine all 36,249 PRIDE datasets released between 2013 and 2025, following the expanding contributor community from experimental scale and recorded workflows to the analytical results shared in deposited files. We then use file dates and literature links

to place deposits in the history of their experiments and examine their subsequent visibility in published work. These observations describe the public record: instrument labels do not establish ownership or adoption dates, and mentions of dataset identifiers in later papers do not by themselves demonstrate reanalysis. Within these limits, the archive allows us to examine how a growing community shares its work and where more of the results already preserved by PRIDE could become discoverable.

Results

PRIDE reached a broader contributor community

PRIDE expanded across contributor groups and the journals publishing their work. Comparing 2023 to 2025 with 2013 to 2015, the effective number of groups increased 5.0-fold and that of journals 5.4-fold (95% interval 4.7 to 6.3). Effective numbers account for both the number of categories and how evenly datasets are distributed among them (Fig. 1a,b; Box 1; Supplementary Tables S2 and S3). The number of distinct journals publishing linked papers also grew (Supplementary Tables S18 and S22).

Broader participation accompanied more even representation of instrument models, whose effective number increased 1.8-fold (1.6 to 2.0). The number of distinct models expected in equal-sized samples did not increase, however, while the effective number of species decreased to 0.67 times its earlier value (0.56 to 0.80). These directions agreed across the two diversity measures that give different weights to common categories. Other dimensions showed no detectable change at order 2, and the timing of increases varied among dimensions (Fig. 1a; Supplementary Tables S3 and S22).

Human and mouse accounted for most of the concentration in species. Datasets naming neither species fell from 46% of deposits during 2013 to 2016 to 33% during 2023 to 2025. Their share also declined among groups contributing in both periods, so changes in which groups contributed explained only part of the decline (Supplementary Fig. 9; Supplementary Table S22).

New contributors continued to supply a substantial share of PRIDE's deposits. Groups with no earlier PRIDE release since 2013 contributed between 42% and 48% of annual deposits during 2020 to 2025 (Fig. 1c,d). Two-thirds of all groups deposited a single dataset, accounting for a quarter of all datasets (Supplementary Tables S4 and S5; Supplementary Fig. 1).

Growth was geographically uneven, with half of the additional country-resolved datasets coming from Europe when comparing 2023 to 2025 with 2016 to 2018. Deposits from China grew faster than those from Germany, increasing China's share of the archive (Fig. 1e to g; Supplementary Figs 10 and 11; Supplementary Tables S6 and S22). These counts concern PRIDE only and exclude groups that deposit only in other repositories, including iProX[9], jPOST[10] or MassIVE[2].

Archive growth encompassed experiments at different scales

As PRIDE expanded[1], the typical deposit contained fewer raw runs, even as large datasets became more numerous. The median number of raw runs (mass-spectrometry acquisitions) per dataset fell from 30 in 2013 to 2016 to 16 in 2023 to 2025, a ratio of 0.53 (95% interval 0.48 to 0.63). The whole distribution shifted towards fewer runs (Fig. 2a; Supplementary Table S8).

Despite this decrease in the median, the mean annual number of datasets with at least 100 runs rose from 124 to 459 across the same periods. Their share of deposits nevertheless fell as the archive grew (Fig. 2b; Supplementary Table S22). Such deposits are relevant to the scale proposed for proteomics studies of participant cohorts[6,7]. Run counts differ from sample counts because isobaric labelling measures several tagged samples together in one run. The number of channels per labelled dataset increased as TMTpro[11] replaced iTRAQ[12] (Supplementary Fig. 8; Supplementary Table S22).

The decrease in median runs persisted when we changed counting rules, combined datasets from the same paper, accounted for shifts in deposit composition and stratified by country. It was not detectable in every subgroup, including datasets labelled with tandem mass tags (TMT)[13] or naming SCIEX or Bruker instruments; the early Bruker subset was small (Fig. 2c; Supplementary Tables S8 and S22).

Changes among returning contributors and the arrival of new groups both accompanied the shift towards fewer runs. The decline was shallower within groups contributing in multiple years than in the pooled comparison of those groups. First-time contributors also deposited fewer runs than comparable returning groups: among 2018 to 2025 releases, the matched ratio of geometric means was 0.64 (0.61 to 0.68). Geometric means average counts on a logarithmic scale. The comparison accounted for release year, vendor, acquisition mode, human-sample status and continent; the within-group slope comparison is reported separately in Supplementary Table S22.

Individual runs occupied more storage, with median raw-run size increasing from 0.35 GB in 2013 to 1.07 GB in 2025. In datasets with one identified instrument model and exact run counts, the changing mix of models accounted for the increase in mean log10 bytes per run; the within-model component did not show an increase (Fig. 2d; Supplementary Tables S7 and S22). With fewer but larger runs, median dataset size remained between 19 and 25 GB in every year, while the mean and the sizes of the largest datasets grew.

Storage was concentrated in a small fraction of deposits: the largest 10% held 61% of the archive's 2.71 petabytes. Most storage was occupied by raw files in instrument manufacturers' formats (Fig. 2e,f; Supplementary Fig. 6; Supplementary Table S22).

Deposits recorded technological change alongside established workflows

PRIDE's instrument labels document changing workflows alongside persistent earlier technologies. Thermo Fisher labels accounted for most instrument assignments throughout the study, consistent with the repository's reported share of stored raw files[1]. Successive leading Orbitrap models[14,15,16] reached progressively smaller peak shares. These comparisons give each dataset one count divided equally among its named models (Fig. 3a; Supplementary Fig. 2; Supplementary Table S22).

After combining labels that refer to the same instrument model, the effective number of models rose from 4.7 (95% interval 4.0 to 5.7) in 2013 to 12.1 (11.5 to 12.8) in 2025. Every model named in at least 50 datasets was still represented among 2025 deposits, which include reanalyses and data released after a delay from the experiment (Fig. 3b). Deposits naming timsTOF models[17] grew alongside this persistence of older instruments, consistent with reports of the family's increasing use[18] (Supplementary Table S22).

New models appeared in deposits from Asia about one year later than expected from its dataset count. Africa's longer delay was close to that expected from its small number of datasets. These expectations came from drawing instrument-model labels independently of continent within each release year, without accounting for similarities among datasets from the same contributor group (Supplementary Fig. 3).

DIA-NN[19] rose from being named in 2 datasets released in 2020 to 722 in 2025, ranking third behind MaxQuant[20] and Proteome Discoverer[21] (Fig. 3c; Supplementary Table S22).

Software naming also varied geographically: in 18 of 20 tool-country pairs, groups in the developers' country named the tool more often than groups in all countries combined. The median ratio was 1.8-fold (permutation P = 0.0002; Fig. 3d). Tool-specific comparisons are provided in Supplementary Table S22, with intervals in Figure 3d. The developers' own laboratories were not excluded. A similar pattern had been reported for open-source analysis workflows in western Europe[21].

Parameter files and logs showed that older software versions continued to appear in new deposits. Among 2025 submissions with a recovered version, the DIA-NN 1.8 series accounted for 61% (48 to 74%). MaxQuant's 1.6 series, first recorded in PRIDE in 2017, also remained common. Recovery coverage and the separate comparisons by submission and release year are given in Supplementary Tables S12 and S22 (Supplementary Fig. 4).

Older measurements were associated with older software versions. Among MaxQuant datasets submitted in 2023 to 2025, versions beginning with 1 were used in 35% of datasets whose newest raw file was less than one year old, compared with 80% when it was more than three years old (Supplementary Table S22).

Tool-specific files revealed analytical content beyond standard formats

The deposited files contained far more recognisable peptide or protein identification results than standard formats alone revealed. Among datasets released in 2025, reading table column headings brought detection of identification results to 80%, compared with 16% containing standard result files (Fig. 4c; Supplementary Table S13). Inspecting archive contents and then table headers progressively recovered results missed by file names alone. Even with all three search stages, the detectable share declined over time (Fig. 4d; Supplementary Table S22). These percentages describe files detected by the recognition rules. Missed files and false matches can affect the estimates; detection alone cannot show whether a file is valid or usable.

Proteomics Standards Initiative formats provide common representations of results; mzIdentML has been implemented in over 35 software tools[8]. We counted submitter-deposited mzIdentML[22] or mzTab[23] as standard result files and excluded repository-generated files (Box 1). Their annual prevalence remained within a narrow range after 2015, despite large differences among analysis tools. DIA-NN had the lowest rate and MS-GF+[24] the highest; Spectronaut and MaxQuant also had low rates, whereas Mascot[25] deposits more often included standard files (Fig. 4a; Supplementary Tables S9, S13 and S22).

Software export capability was associated with what contributors deposited. Standard files appeared in 55% of datasets naming tools that write them by default (95% interval 44 to 66%), compared with 3% (2 to 4%) for tools that cannot write them. Tools offering standard files as an option lay between these groups (Fig. 4b; Supplementary Table S10). The default-writing group contained only MS-GF+ and MetaMorpheus[26], and the latter's low deposition rate shows that default output did not guarantee inclusion in PRIDE (Supplementary Table S22).

Changes in the mix of analysis tools helped explain the archive-wide pattern. Between 2019 to 2021 and 2023 to 2025, the standard-file rate changed by −1.4 percentage points (−3.0 to 0.3). A shift towards software families with lower deposition rates, mainly DIA-NN and Spectronaut, offset improvements within families; Mascot's share declined over the same period. Supplementary Table S22 reports the two components and their intervals.

Data-independent acquisition (DIA), which fragments precursor ions within predefined mass-to-charge windows[27], became more common while standard result files remained uncommon in DIA deposits. The standard-file rate in datasets with no DIA detected stayed relatively stable (Supplementary Table S22). Differences in spectral-library formats, which store collections of reference spectra, have been described as one obstacle to DIA reanalysis[28].

The contents of results archives also distinguished intermediate processing from final outputs. Files containing scan-level information and intermediate search outputs occupied much more space than final result tables (Supplementary Fig. 6). Deposits containing spectra in open formats became less common, partly reflecting a decline in submissions classified as complete, a category requiring lists of measured spectral peaks (Supplementary Fig. 7; Supplementary Table S22).

Documentation was uncommon but often recurred within contributor groups. When a group's previous dataset included a README, its next did so in 41% of cases, against an expected 1.5% for datasets released in the same years. Sample and data relationship format (SDRF) files[29], which link sample descriptions to data files, showed a similar pattern (Fig. 4e; Supplementary Table S14). These recurrence rates are unadjusted for pairs submitted on the same day or for the same paper, and some SDRF files were added to several datasets of a group after release. SDRF presence increased over time but remained limited[30] (Supplementary Table S22).

Release dates placed public deposits in the history of an experiment

Dates preserved in deposited files connected experiments and analyses to public release. Raw-file names supplied a measurement date for 34% of datasets with raw files; in this subset, the median interval from the latest measurement to public release was 2.2 years (95% interval 2.1 to 2.3) (Fig. 5a; Supplementary Tables S15 and S22).

Deposited files also preserved the ages of resources used in analysis. Sequence-database dates and the first recorded appearances of software releases generally preceded public release by several years (Supplementary Fig. 5; Supplementary Tables S11, S15 and S22). Among datasets with both relevant dates, these resources often predated even the latest measurement. Software ages are lower bounds because a version may have existed before its first recorded PRIDE appearance.

Most of the interval from measurement to public release elapsed before submission. The median interval from submission to publication of the linked paper was 141 days, whereas the median delay from publication to public release was 25 days (Supplementary Table S15). Releases also clustered on particular days, and datasets released on those days had longer delays after publication. The comparisons and definition of these days are given in Supplementary Fig. 13 and Tables S16 and S22. Who set each release date is unrecorded; Supplementary Fig. 14 shows the separate record of submission timestamps.

PRIDE-linked publications reached beyond specialist journals

PRIDE-linked papers reached an increasingly broad range of journals. Journals with titles containing "proteom" published 51% of linked papers in 2013 and 10% in 2025, even as the number they published increased. The leading journals also accounted for a

smaller share of the total (Fig. 5b,c; Supplementary Tables S17, S18 and S22). In 2025, *Nature Communications* published the most PRIDE-linked papers of any single journal.

The research fields of linked papers shifted from chemistry towards medicine. Almost all of the chemistry category was spectroscopy, the subfield to which mass-spectrometry methods papers are assigned (Supplementary Fig. 12; Supplementary Table S22). Coverage limits these comparisons: paper links were available for 94% of datasets released in 2013 but only 61 to 67% of those released since 2021.

Accession mentions provided evidence of continued attention to public data

Text mining in Europe PMC[31] identified papers that named PRIDE dataset identifiers, called accessions. We considered mentions in papers other than the dataset's primary reference that named deposits from multiple contributor groups, applying the eligibility criteria in Methods. These papers mentioned 2.3% of datasets released in 2013 to 2022 within the release calendar year or the next three calendar years (Fig. 5d; Supplementary Table S19). Within a calendar year, we could not determine whether the paper appeared before or after public release. This measure gives an incomplete view of reuse, particularly when accessions appear only in supplementary tables or literature outside open-access full text. A mention does not establish reanalysis.

Papers meeting these criteria named DIA datasets and datasets with human samples more frequently. The adjusted odds of a mention were higher for DIA datasets (odds ratio 1.71; 95% interval 1.19 to 2.39) and for datasets with human samples (1.39; 1.14 to 1.70). The estimate for datasets assigned only to Thermo Fisher was above one, but its interval included one, so an association was not established (Fig. 5e; Supplementary Table S20). The model accounts for the other dataset and study characteristics specified in Methods; its DIA cohort includes releases only up to 2022.

We did not detect a relationship between standard result files and qualifying mentions (adjusted odds ratio 0.89; 0.66 to 1.19). Extending the mention window or restricting the comparison to a more similar set of datasets did not establish an association, although the restricted comparison was imprecise. The populations, windows and unadjusted rate ratios for these sensitivity analyses are reported in Supplementary Table S20. A modest advantage in mention rates for datasets with standard result files therefore cannot be excluded.

For SDRF files, the timing of their addition complicates the association with later mentions. Datasets with SDRF files had higher odds of a qualifying mention (odds ratio 4.08; 2.38 to 6.65), but many were released before the format existed and acquired these files later, often through reanalysis projects (Supplementary Table S22). The direction of the association therefore cannot be determined. Methods papers, machine-learning papers and papers presenting research resources were among those naming the largest numbers of datasets.

Discussion

Thirteen years of PRIDE deposits document the work of an expanding proteomics community. New contributors continued to account for a substantial share of annual releases, and linked papers appeared across a much wider range of journals. This collective record, built by the repository and its contributors, makes changes in research practice visible at a scale difficult to achieve through individual studies.

Examining deposited files shows which experiments contributed to PRIDE's growth. PRIDE accommodates increasing numbers of large deposits alongside many datasets with fewer raw runs. The mix of instrument models helps explain why a typical run occupies more storage. Older models remained represented in deposits, which include delayed releases and reanalyses. Deposited software versions also preserve evidence of earlier workflows. Participation can broaden even as biological coverage becomes more concentrated on human and mouse.

Identification results could be recognised in 80% of 2025 deposits when files from individual analysis programs were examined, although 16% contained standard result files. Recognising these outputs more systematically could make contributors' results easier to discover. File presence does not establish validity or readiness for reuse, so extraction would need validation and clear records of how the files were produced and processed.

Differences in deposition rates across analysis programs suggest where improvements could be tested with tool developers. Standard result files accompanied 55% of datasets whose tools write them by default; for tools lacking this capability, the rate was 3%. The low rate of standard result files for MetaMorpheus, which writes these files by default, shows that a default alone is insufficient, and tool choice may reflect depositor intentions. Sample annotations supplied during analysis could be used to create SDRF files, which link samples to data files, and reduce reconstruction work for later users[29,30]. Recording the instrument, software version and database date from deposited files would help readers trace how the data were produced, a concern addressed in reporting recommendations for single-cell proteomics[32].

Release timing helps explain which stages of a study PRIDE records. Most of the measured time from data acquisition to public release elapsed before submission, so it cannot be attributed wholly to repository handling. Older software often accompanied older measurements, so a public release date needs to be read alongside the available acquisition and analysis dates.

Mentions of dataset identifiers in qualifying papers other than the dataset's primary reference provide an incomplete view of attention to deposits. These papers more often named DIA and human datasets. Standard result files showed no detectable association with those mentions, but the intervals do not exclude a modest benefit. Mentions alone do not establish reanalysis. Calendar-year mention windows cannot show whether a paper appeared before or after a dataset's release within the same year. The SDRF association also requires caution because many files were added after release, sometimes during reanalysis. Evaluations of future improvements should record when changes occur and distinguish subsequent use from metadata added during that use.

The scope and coverage of the evidence constrain these conclusions. PRIDE represents only part of ProteomeXchange, which matters especially for geographical comparisons. Identifiers derived from principal-investigator contacts approximate contributor groups: changed addresses can create apparent new groups, while core facilities serve several laboratories. Because the record starts in 2013, a group classified as first-time may have contributed before that year. File and text searches provide incomplete evidence of software, acquisition modes, labelling and identifiable results. Their sensitivity and false-positive rates were not independently measured. Paper links cover 75% of datasets overall but 61 to 67% of recent releases; dated raw-file names cover 34% of deposits containing raw files. Text mining also misses later mentions, particularly those confined to supplementary material. The reconstructed snapshot cannot recover the history of metadata edits or withdrawn datasets and excludes recent

submissions that remain private. Our comparisons are descriptive; the Supplementary Tables give denominators and intervals where computed, and Table S21 records the sensitivity analyses.

Making the analytical results already shared by PRIDE's contributors easier to find and documenting how those results were produced would extend the value of that work.

Box 1. Data, definitions and limits

Cohort. The 7 April 2026 PRIDE snapshot contains 36,249 public datasets with release dates between 2013 and 2025 (Supplementary Table S1). Release year means the year when a dataset became public in PRIDE. We state explicitly when an analysis uses submission year instead.

Units. Results are reported per dataset unless stated otherwise. Papers are publication records in OpenAlex[33], a catalogue of scholarly works, matched to the publication records of PRIDE datasets. Contributor groups are defined by hashed contacts of principal investigators. A first-time group has no earlier PRIDE dataset released since 2013. For diversity calculations, a dataset naming several instrument models, tools or species contributes equal fractions of its credit to those categories. Species-presence percentages in Supplementary Figure 9 instead count a dataset once for each species named, so those percentages can sum to more than 100%. Other denominators are specified in the figure legends and tables.

Effective number. Hill numbers[34] express diversity as the number of equally common categories that would have the observed diversity. They are calculated from category shares p_i among datasets. We use $1/\Sigma p_i^2$ at order 2, which gives more weight to common categories. At order 1 the expression is $\exp(-\Sigma p_i \ln p_i)$. Order 0 measures richness, the number of distinct categories. We rarefied richness by comparing equal numbers of datasets[35].

File evidence. We examined top-level file lists (3.03 million files) and the members of 45,994 results archives. Archive members were listed from zip central directories, which index the files in an archive, using HTTP requests for selected byte ranges. We matched column headers in deposited tables to characteristic outputs of analysis tools. We also extracted software versions from parameter files, logs and workflow files, and dates from raw and database file names. Each source provides incomplete evidence of file content. Recognition rules can miss results or produce false matches; detected shares are not validated estimates of the true prevalence of usable results.

Standard result files. These are submitter-deposited mzIdentML or mzTab files, including files identified by expanded name matching; files generated by the repository are excluded.

Uncertainty. We repeatedly sampled whole contributor groups with replacement to obtain bootstrap distributions. Their percentiles supplied 95% intervals (seed 20261001; 200 to 2,000 replicates depending on the analysis). Contributor-group effective numbers are descriptive, without bootstrap intervals. Mention windows use calendar years and do not resolve within-year chronology.

What the data cannot show. The records do not establish instrument ownership or purchase dates, laboratory size, the reasons for packaging decisions, who set a release date, or reuse that leaves no text-mined mention. Country is unresolved for 22% of datasets.

Methods

Data sources. The analysis used frozen snapshots documented in `data/early/Data-manifest.json` and `data/reconstruction/Source-manifest.json`. Sources were PRIDE archive tables (projects, files, instruments, species and publications), ProteomeXchange registry links, OpenAlex records of linked papers (works, authorships and institutions) and Europe PMC records (journal titles and accession mentions).

Reading of files. We listed the files inside zip archives, including zip64 archives, by reading their central directories with requests for selected byte ranges. We read column headers from the first bytes of deposited text and xlsx tables and matched them to characteristic column patterns in the outputs of analysis tools. We extracted software release strings from parameter files and logs. Archive members were classified with the same name-based rules as top-level files. For database-age analyses, the largest dated FASTA represented each dataset; equal-sized candidates were ordered by file name and location. The header patterns used for recognition underwent model-assisted review and revision. This review did not provide an independent estimate of classification accuracy. Recognition of a file type does not verify its contents or usability. A repeat retrieval of the frozen targets on 11–12 October 2026 recovered 46,147 archive listings and 59,227 table-header records. Reclassification changed the final recognition outcome for one dataset; all annual recognition percentages agreed with the frozen analysis at one-decimal precision. The current analysis uses the repeat retrieval, while the release retains the earlier derived inputs and a comparison report so that these changes can be checked.

Runs. Raw runs are mass-spectrometry acquisitions stored in vendor raw formats and identified from file names and extensions. Because bundled formats were counted as single runs, these counts are lower bounds; exact counts are reported separately.

Instrument models. We harmonised instrument labels to model names. We excluded labels naming only a series or vendor, or no instrument, from model-level analyses.

Statistical uncertainty. We obtained 95% percentile intervals by bootstrapping contributor groups. For categories other than contributor groups, we calculated period ratios of effective numbers by resampling contributor groups independently within each period. Effective numbers of contributor groups and their period ratios are descriptive point estimates without bootstrap intervals. Resampling groups while retaining their category identities biases this concentration statistic.

Comparisons. Shift-share decompositions separate changes within categories from changes in the proportions of datasets in those categories. Rates of qualifying accession mentions were directly standardised: we weighted rates within release-year × human-status strata to match their shares in the full cohort. The expected first appearance of instrument models by continent was obtained by drawing model labels independently of continent within each release year.

Accession-mention outcome and model. For datasets released in 2013–2022, we modelled whether a qualifying non-primary paper mentioned the accession in the release calendar year or the next three calendar years. Qualifying papers named deposits from at least two contributor groups and no more than 100 datasets. We could not determine whether a same-year mention preceded or followed the dataset's release. A binomial logistic model estimated the probability of a qualifying mention. It used a ridge penalty (10^{-4}), which shrinks coefficients towards zero, and included standard result files, detected data-independent acquisition (DIA), sample and data relationship format (SDRF) files, human samples and a Thermo Fisher-only assignment from instrument labels. The model also adjusted for categorical release year, grouped software family and research field, log(1 + citations), a missing-citation indicator and $\log_{10}$(dataset bytes). The model included 20,684 datasets, of which 483 had a qualifying mention, from 8,363 contributor groups. Percentile intervals used 2,000 resamples of whole contributor groups. Supplementary Methods specify the transformations, treatment of missing values and reference levels. Sensitivity analyses reported unadjusted rate ratios for a restricted cohort with no DIA detected. A separate sensitivity analysis extended the mention window to the release calendar year plus five subsequent calendar years.

Sensitivity analyses. Each principal result was re-examined using alternative definitions, units of analysis and subsets. The outcome of each check (supported, weakened or refuted) and the resulting word-

ing are recorded in Supplementary Table S21.

Computational implementation. We queried the data with DuckDB[36] and analysed them in Python using NumPy[37], pandas[38] and SciPy[39]. The accession-mention model used statsmodels[40], and the data figures were drawn with Matplotlib[41]. The supplementary reference guide identifies the analysis programs, instrument models and file formats named in the text and figures.

Figures. Main and supplementary data figures were generated from code using frozen analytical outputs. Their captions state the relevant selection criteria, denominators and quantities estimated. Raster exports of the main and supplementary data figures have a width of 7.20 inches at 600 dpi. The unnumbered BioRender overview uses its native 2,752-pixel-wide preview, displayed at 7.2 inches (approximately 382 dpi).

AI assistance. OpenAI Codex assisted with code development and review, figure preparation, literature checking and manuscript drafting and editing. AI assistance does not provide independent validation of the scientific claims.

Supplementary information

Supplementary Figures. 1, Concentration of datasets and bytes across contributor groups. 2, History of every common instrument model. 3, First appearance of new instrument models by continent. 4, Software version series in deposited files. 5, Age of software releases and sequence databases. 6, How the archive's bytes are distributed among file types. 7, File formats. 8, Isobaric labelling and post-translational modifications. 9, Organisms. 10, Countries. 11, International co-authorship. 12, Research fields of linked papers. 13, Timing of public release. 14, Submission timestamps.

Supplementary Tables S1 to S22. Supplementary-Tables.xlsx contains the tables; its contents sheet gives each table's definition and source. Table S22 collects supporting numerical comparisons, including their populations, periods and reported uncertainty.

The Supplementary Methods explain how mentions were identified, how the model was constructed and which inferences it supports.

Data availability

The source metadata, derived inputs, figure evidence and supplementary tables are distributed with the PRIDE data-economy repository. Source-metadata.zip contains the frozen source tables; Extraction-caches.zip contains the recovered archive listings and table headers. Public-analysis-release.zip contains the manuscript, derived tables, code and validation records, and Figures-600dpi.zip contains the 19 numbered figure masters. Table S22 is a curated summary of supporting comparisons at manuscript reporting precision. Checksums identify the source metadata and the archive listings and table headers retrieved on 11–12 October 2026. Software-version observations and flags from table headers inside zip archives remain extracted inputs; their original response bytes are unavailable. Contributor labels are pseudonymised, and the mapping back to contributor identities is excluded. The repository documents these boundaries and compares the rebuilt results with the earlier analysis. Author-created data, text and figures use CC BY 4.0, excluding third-party assets such as the BioRender graphical abstract. The Zenodo deposit is being prepared; the release status is recorded in the repository.

Author declarations

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Funding. This research received no specific funding.

Competing interests. The author declares no competing interests.

Author contributions. Filip Rumenovski conceived and directed the study and developed its scientific scope and interpretation. AI assistance with analysis, code, figures and manuscript preparation is described in Methods.

Code availability

Custom analysis and figure-generation code is available under the MIT licence at github.com/filiprumenovski/pride-data-economy. The offline pipeline reconstructs geography assignments, file classifications, run-count and timing sensitivity analyses, the adjusted mention model, all 19 numbered figures and 22 supplementary tables. The repository provides the pinned environment, checksummed inputs and reproduction instructions. Software-version and ZIP-contained-header extraction boundaries are described in Data availability and the repository documentation.

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Main figures

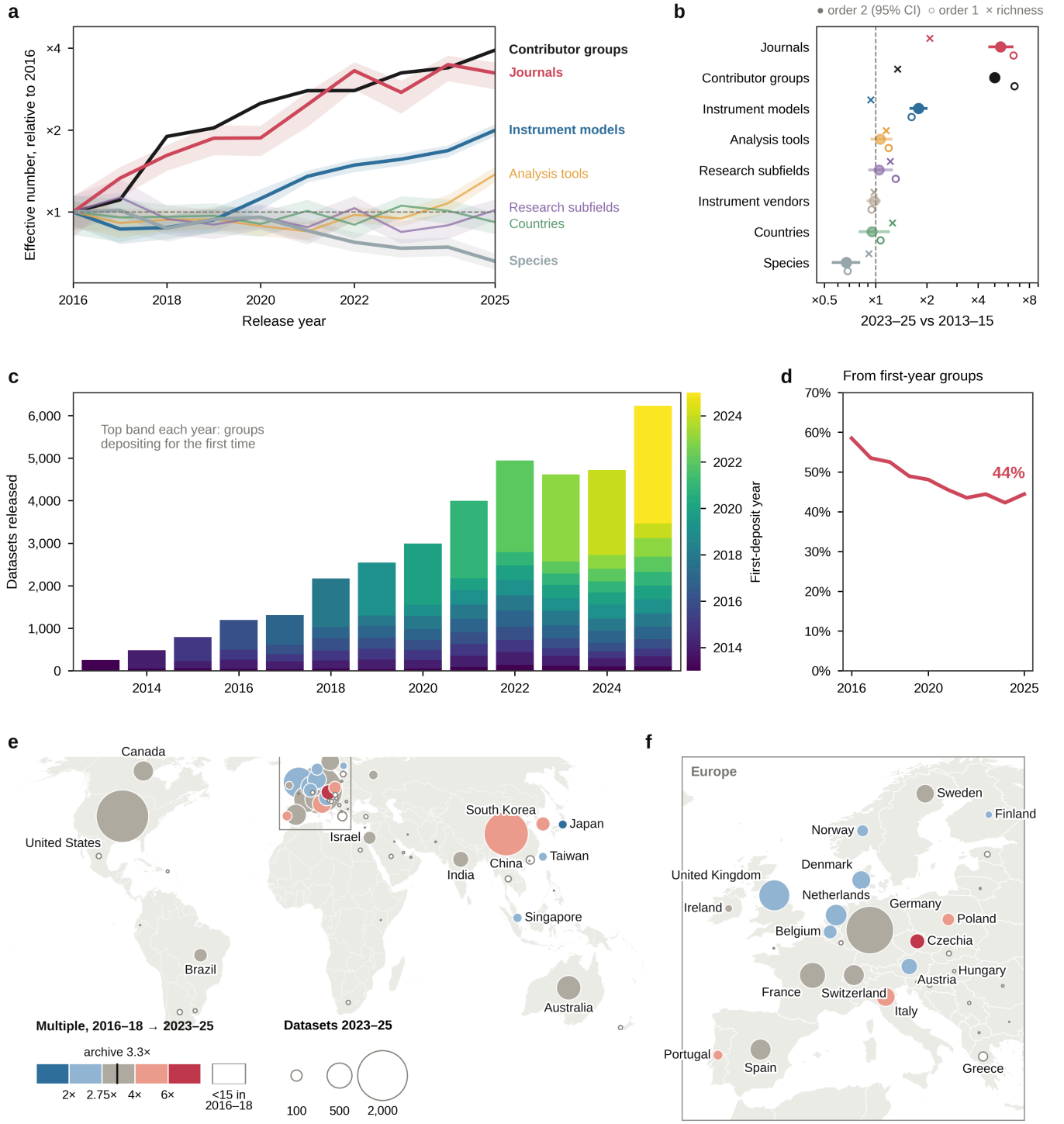


Figure 1 | An expanding community contributing to PRIDE. a–b, Diversity increased among contributor groups (×5.0), journals (×5.4) and instrument models (×1.8), and decreased among species (×0.67). Effective number (Hill number) combines the number of categories with their relative shares among PRIDE datasets. a, order 2 (inverse Simpson) by release year relative to 2016. b, 2023–25 vs 2013–15 at orders 2, 1 and 0 (category counts adjusted to equal dataset counts); 95% intervals from resampling contributor groups, except for contributor-group diversity, which is descriptive. A dataset naming several categories contributes equally to each. **c–d,** Four in ten public datasets come from groups depositing for the first time. PRIDE datasets per release year, grouped by the year the contributor group first deposited; groups are defined from hashed principal-investigator contacts. **e–g,** Europe supplied half of PRIDE's 3.3-fold growth; dataset counts grew 5.2× for China and 2.8× for Germany. Circle area: datasets released 2023–2025, by the contributing group's country; colour: multiple of the 2016–2018 count (archive-wide 3.3×). 3,632 and 12,085 datasets with an assigned country (78% of each period; the rest have no country assignment). Datasets added: Europe 4,218 (50%), North America 1,852 (22%), Asia 1,848 (22%; China 1,283). Counts describe deposits, not instrument ownership or national research output.

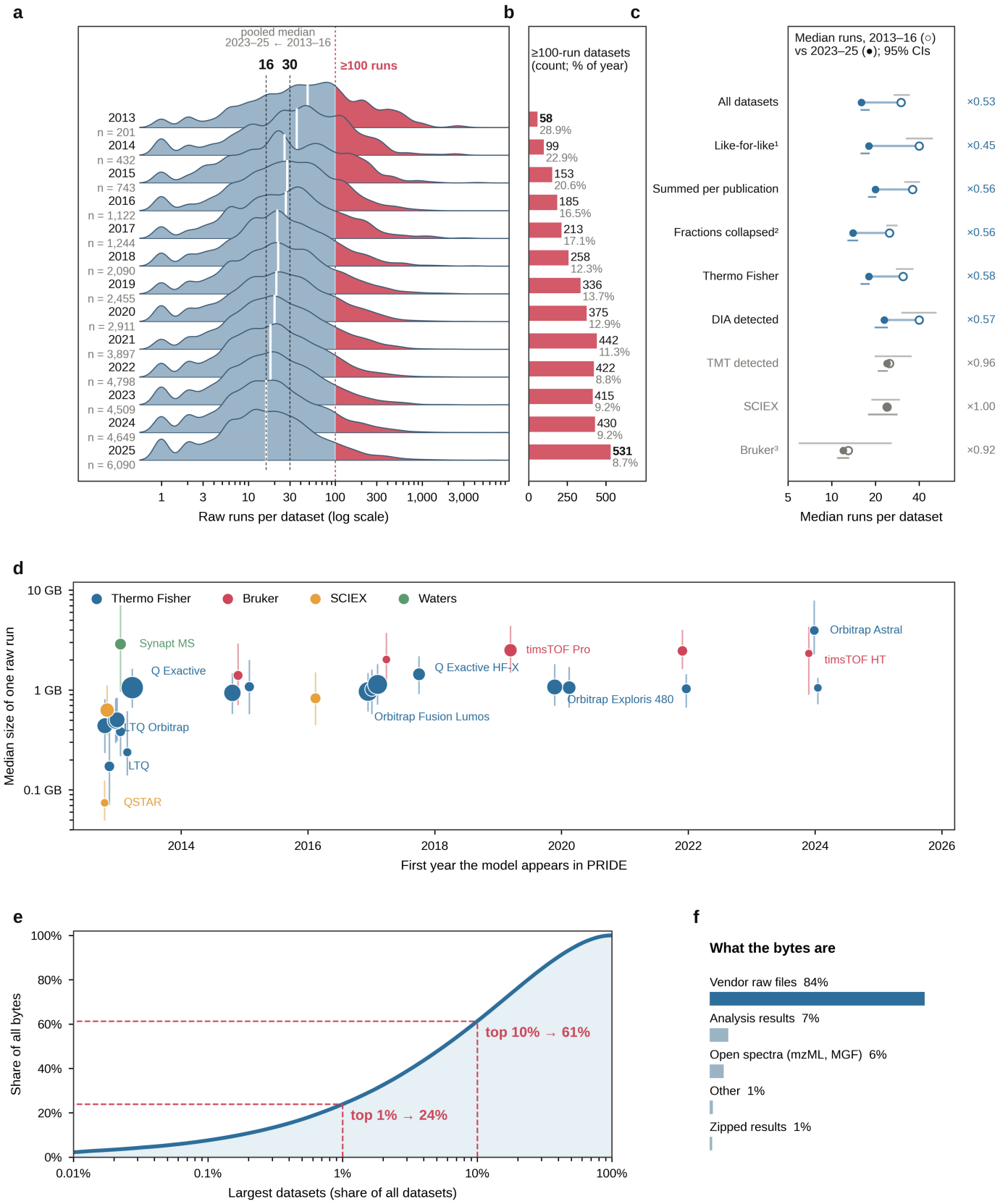


Figure 2 | A growing archive spanning different experimental scales. a–c, Typical datasets hold half as many raw runs as in 2013–16, yet the number with ≥100 runs rose from 58 in 2013 to 531 in 2025. PRIDE datasets released 2013–2025 with ≥1 identified raw run or raw bundle (n = 35,141 of 36,249). Runs are instrument acquisitions stored in vendor raw formats; bundled files make these counts lower bounds. Runs are not samples: TMTpro-18plex tandem mass tags can combine 18 samples in one run. Ridges: distribution per release year (equal area, log axis); white tick = median. Pooled medians 30 (2013–16, n = 2,498) and 16 (2023–25, n = 15,248); 95% intervals from resampling groups defined by hashed principal-investigator contacts. ¹ Thermo Fisher, no detected data-independent acquisition (DIA) or TMT, human samples, linked publication. ² Runs with fraction numbers combined into one count as an approximation to sample count. ³ Bruker 2013–16 n = 56. Grey: the confidence interval for the ratio includes 1. **d,** Median raw-run size is 9× larger for the Orbitrap Astral than for the LTQ Orbitrap. Median size of one raw run by instrument model (bars: interquartile range), placed at the model's first year in PRIDE. **e–f,** 1% of datasets hold 24% of the archive's 2.71 petabytes. Cumulative share of PRIDE bytes held by the largest datasets (all deposited files, 2013–2025).

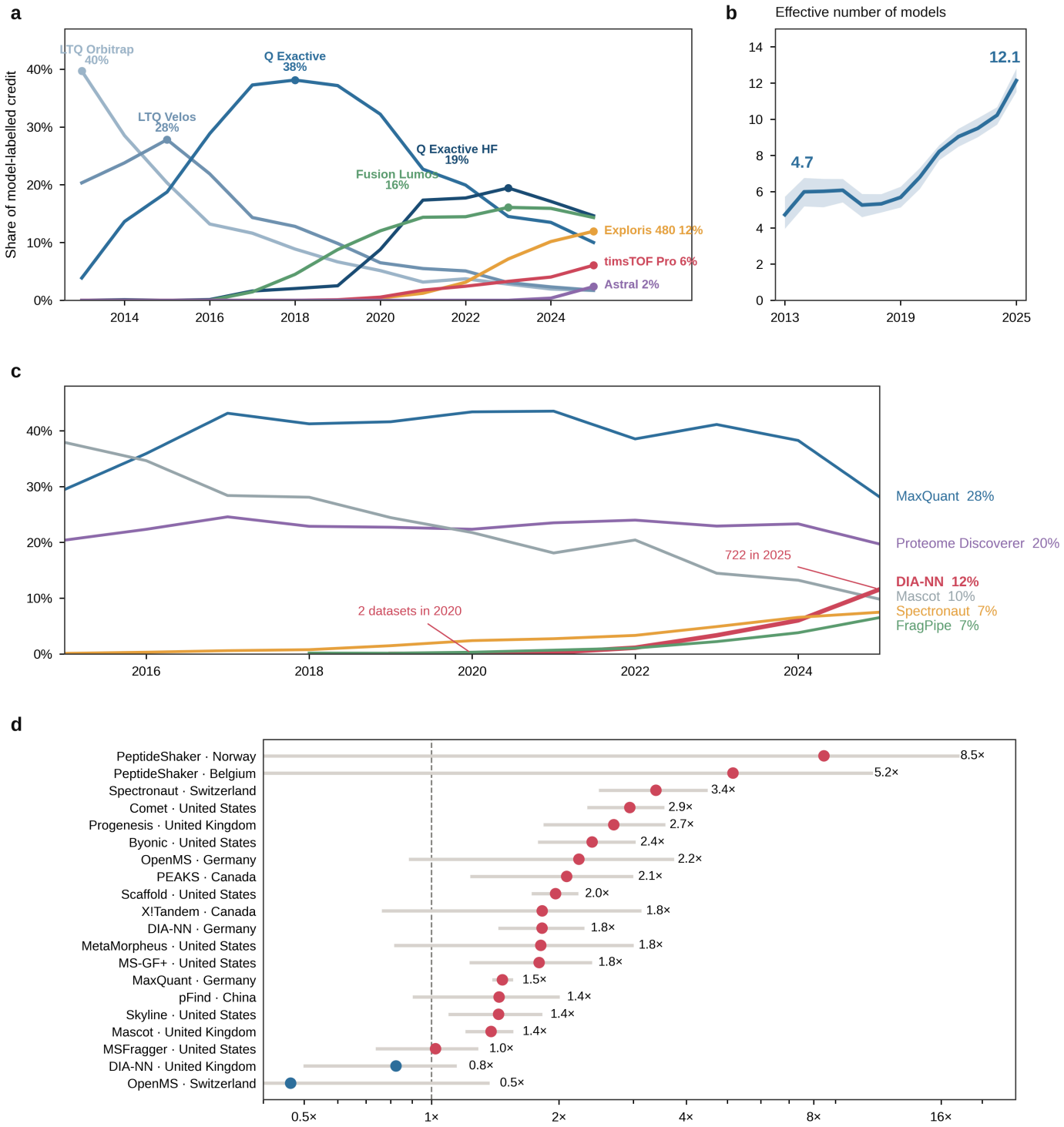


Figure 3 | A public record of instruments and analysis workflows. a–b, Deposits named new instrument models alongside older ones. a, Share assigned to each selected model among datasets with an instrument-model label; a dataset naming several models contributes equally to each. Every model named in ≥50 datasets still appeared in 2025 releases; deposits include delayed releases and reanalyses. b, Effective number of models (inverse Simpson index of the weighted model shares); shading: 95% intervals from 300 bootstrap resamples of contributor groups. **c,** Within five years, DIA-NN became the third most frequently named analysis program in public datasets. Share of PRIDE datasets naming each analysis tool (protocol text or software field), by release year. **d,** Software is named more often in deposits from its developers' country: 18 of 20 tool–country pairs. Frequency of each program among groups in its developer's country relative to all countries combined (1 = equal frequency); 95% intervals from resampling contributor groups. Median 1.8×; permutation p = 0.0002.

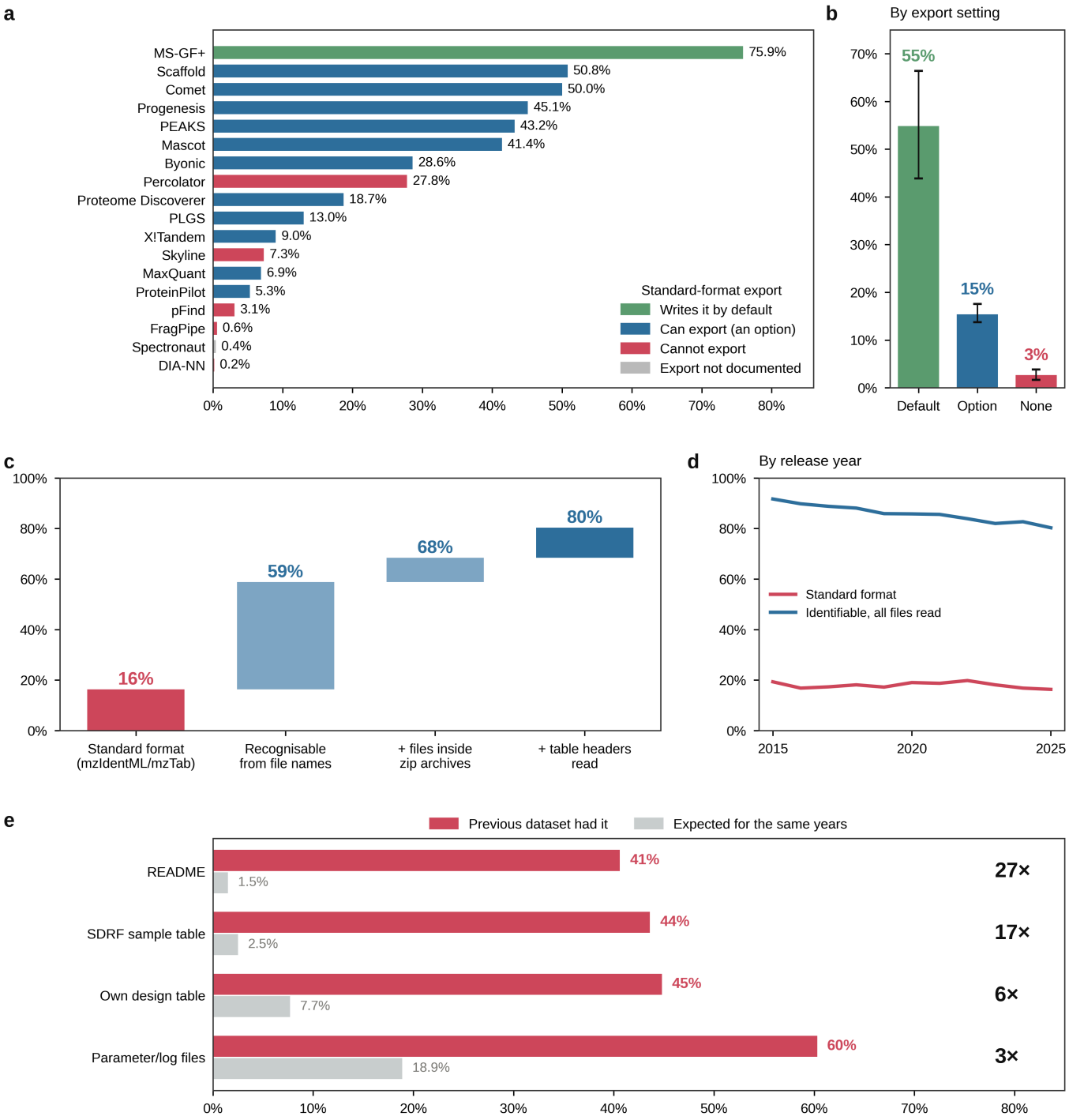
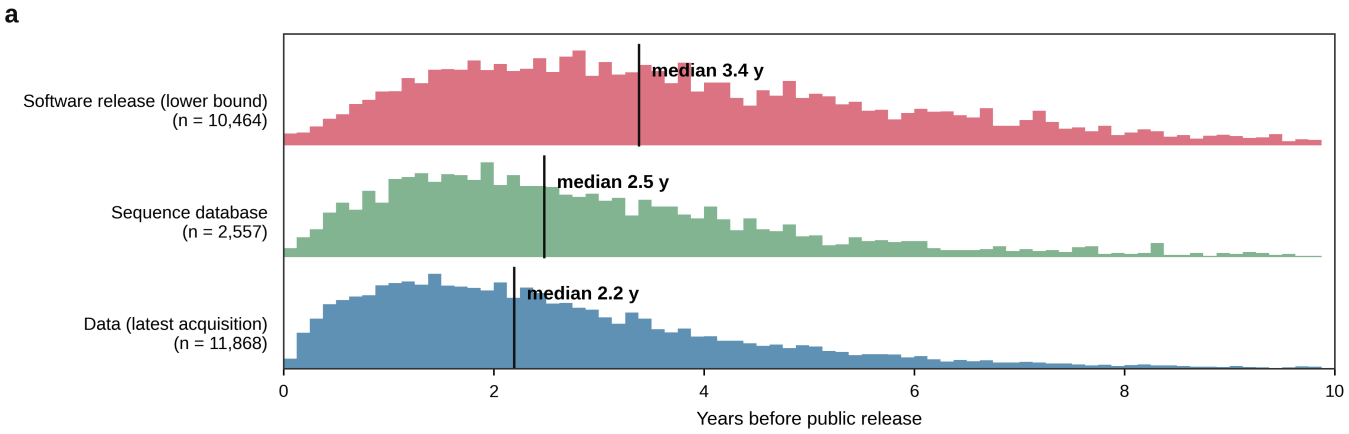
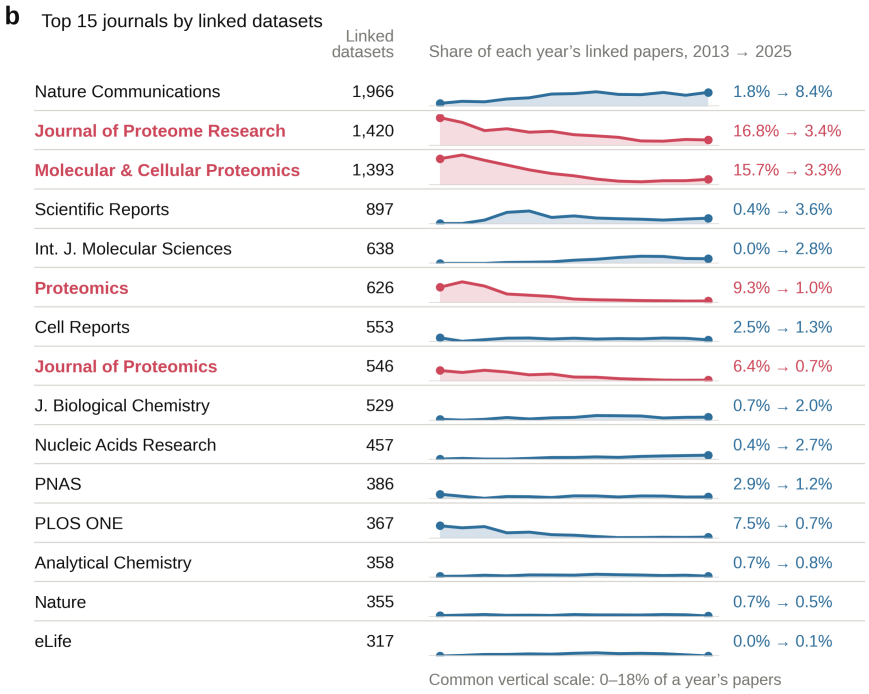
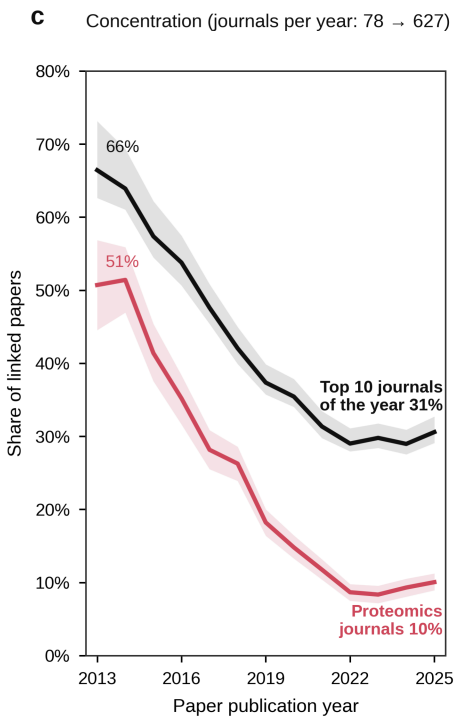
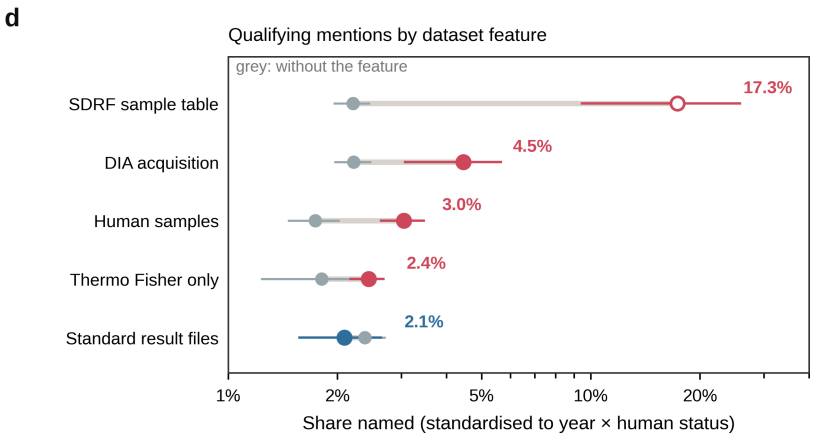


Figure 4 | Results shared in standard and software-specific formats. a–b, The share of datasets with standard result files ranges from 0.2% to 76% across analysis programs. Denominator: all eligible PRIDE datasets naming a single analysis program; programs require ≥60 datasets. Numerator: those containing submitter-deposited mzIdentML or mzTab files. Repository-generated files are excluded. **c–d,** Reading deposited outputs reveals identifiable results in 80% of 2025 datasets; 16% contain standard result files. PRIDE datasets released in 2025 (n = 6,231) assessed for identification or quantification results using file names, archive contents and table headers. Percentages describe rule-based detection; sensitivity and false-positive rates were not independently measured. **e,** README documentation and sample-to-file mappings in SDRF files recur within contributor groups more often than expected from their overall frequencies. Share of a group's next PRIDE dataset carrying each file when the previous one did, vs the rate expected for datasets of the same years.









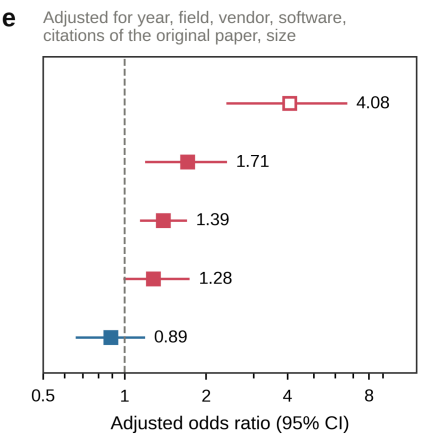


Figure 5 | Connecting public deposits to the research literature. a, At public release, the median time since acquisition was 2.2 years and the software release was at least 3.4 years old. Age at public release of the latest raw-file acquisition date (datasets with dated raw-file names, 34% of those with raw files), the sequence database date and the software release (first seen in PRIDE; lower bound). The plot excludes one negative database interval and two negative software intervals; ages above 10 years are pooled in the final bin. **b–c,** Proteomics journals carried 51% of dataset-linked papers in 2013 and 10% in 2025. Journals of linked papers, obtained from Europe PMC using PubMed identifiers: 22,475 papers published 2013–2025 (525 without a journal, mostly preprints, excluded). Journals ranked by linked datasets; small line plots show each journal's share of papers per year, counting each paper once. Proteomics journals = title contains "proteom" (18 titles). Bands: 95% bootstrap over contributor groups. **d–e,** Qualifying papers more often name human datasets and datasets with data-independent acquisition (DIA); standard result files show no detectable association. PRIDE datasets released 2013–2022 (n = 20,684) mentioned by identifier in a qualifying paper other than their primary reference, during the release calendar year or the next three calendar years (2.3% overall). Qualifying papers name datasets from ≥2 contributor groups and no more than 100 datasets; publication and release order within a year is unresolved. Thermo Fisher only means a single-vendor assignment from instrument labels. Rates are standardised to release year and human status; odds ratios use the adjusted model in Methods, with 2,000 contributor-group bootstrap draws. Europe PMC text searches miss some dataset mentions, and a mention does not establish that the data were analysed again. Open circle: SDRF files were often added after release by reuse projects (91 of 205 datasets predate the format), so the association's direction cannot be determined.

Supplementary methods

Mention ascertainment. We excluded pairs of PRIDE accessions and PubMed identifiers (PMIDs) that were marked as primary references. The main outcome required a mentioning paper to name no more than 100 datasets spanning at least two contributor groups. We computed these counts for each paper before restricting mentions to publication year minus release year of 0–3 inclusive. A paper was not excluded solely because it was the primary paper for another deposit from the same contributor group. Same-year mentions can precede the exact release date. The sensitivity analysis with five subsequent calendar years used publication year minus release year of 0–5 among 2013–2020 releases (11,741 datasets). The restricted cohort comprised 4,399 human datasets assigned solely to Thermo Fisher from instrument labels, with no DIA detected, naming MaxQuant or Proteome Discoverer. Neither restriction establishes raw-file presence or confirms data-dependent acquisition (DDA), in which precursor selection depends on signals in the acquired survey spectra.

Model construction. We fitted a binomial generalised linear model (GLM) with an intercept in statsmodels[40], using its logit link and `fit_regularized(alpha=1e-4, L1_wt=0)`. The logit link relates the predictors to the log odds of a qualifying mention. Missing primary-publication citation counts (5,556 datasets) were replaced with zero before the natural-log transformation log(1 + citations). A separate indicator recorded which citation counts were missing. This indicator does not prove absence of a publication. For each dataset, the citation count was the maximum count reported by iCite[42], the citation-data source, across its linked publication records. Listed byte counts below 10^6 bytes were set to 10^6 before the $\log_{10}$ transformation. Missing values would also be set to this floor. No size values were missing in the model cohort, and 25 fell below the floor. Missing research fields were coded as "None" (4,460 datasets). The eight most frequent resulting categories were retained; all others were pooled as "Other". The seven most frequent software-family categories were retained and the remainder pooled. Datasets naming multiple major tools formed a distinct software group. The reference levels were 2013 for release year, Mascot for software family, and Agricultural and Biological Sciences for research field. Human-sample and SDRF flags were set to false when those features were not detected. Missing contributor-group identities would receive accession-specific identifiers, although this fallback was not needed in the model cohort. All 20,684 dataset rows entered the model's design matrix of predictor values, which had 33 columns including the intercept.

Intervals and estimands. The logistic-model point estimates used every dataset in the cohort. Two thousand cluster-bootstrap fits resampled whole contributor groups with replacement, keeping the same predictor columns and coding. We exponentiated the coefficients to obtain odds ratios and used their 2.5th and 97.5th percentiles as interval endpoints. The seed was 20261001; the source script shares its random-number stream with preceding analyses, so the execution order affects the exact bootstrap sequence. The full-cohort fit and all 2,000 cluster-bootstrap fits returned finite coefficients without fit warnings. Maximum absolute penalised scores, which measure the remaining gradient of the penalised objective, were below 1.0×10^{-5}. The Thermo Fisher-only interval included one in the full bootstrap and in both halves of a split-half diagnostic. This diagnostic assesses Monte Carlo sensitivity; it does not establish interval validity. Figure 5d shows rates directly standardised over release-year × human-status strata to the full cohort distribution, with 1,000 contributor-group bootstrap draws. Strata with no datasets in a subset or bootstrap replicate were omitted. The weights of the remaining strata were rescaled to sum to one. Figure 5e shows adjusted odds ratios, whereas the sensitivity analyses report unadjusted ratios of the mention rate in datasets with each feature to the rate in datasets without it, using 1,000 contributor-group bootstrap draws.

Descriptive diversity. Contributor-group effective numbers are reported without intervals. In the original bootstrap, repeated draws of a group retained the same category identity, changing category concentration. Effective numbers for other categories use contributor-group bootstrap intervals. Bootstrap sampling and rarefaction use a fixed ordering of groups, datasets and categories, so database row order does not change the random draws. The instrument richness comparison rounds to 62 versus 66.

Supplementary reference guide

These references identify the software, instruments and formats named in the text, figures and software registry. They provide technical background and credit to their developers. Dataset counts, shares and comparisons come from this study. A reference to a software family does not identify the version in every deposit or establish its export capabilities. The instrument table retains the recorded model labels and states where a source covers only a family.

Analysis software

Software or grouped label	Reference; qualification where needed
MaxQuant	20
DIA-NN	19
Proteome Discoverer	21 Platform overview.
Spectronaut	43
Comet	44
Mascot	25
MS-GF+	24
MetaMorpheus	26
FragPipe / MSFragger	45,46 Platform and search-engine references; the platform paper is later than the study period.
Skyline	47
PEAKS	48
Scaffold	49
pFind / pLink	50,51 Separate references for the two named programs.
OpenMS / OpenSWATH	52,53 Separate references for the two named programs.
Byonic	54
X!Tandem	55
PeptideShaker / SearchGUI	56,57 Separate references for the two named programs.
SEQUEST	58
ProteinPilot	59 Paragon search algorithm used in ProteinPilot.
Percolator	60
Progenesis	61 Proteomics software family; official documentation.
PLGS (ProteinLynx Global SERVER)	62 Manufacturer documentation.
Sage	63
EncyclopeDIA	64
TPP (Trans-Proteomic Pipeline)	65

Instrument models

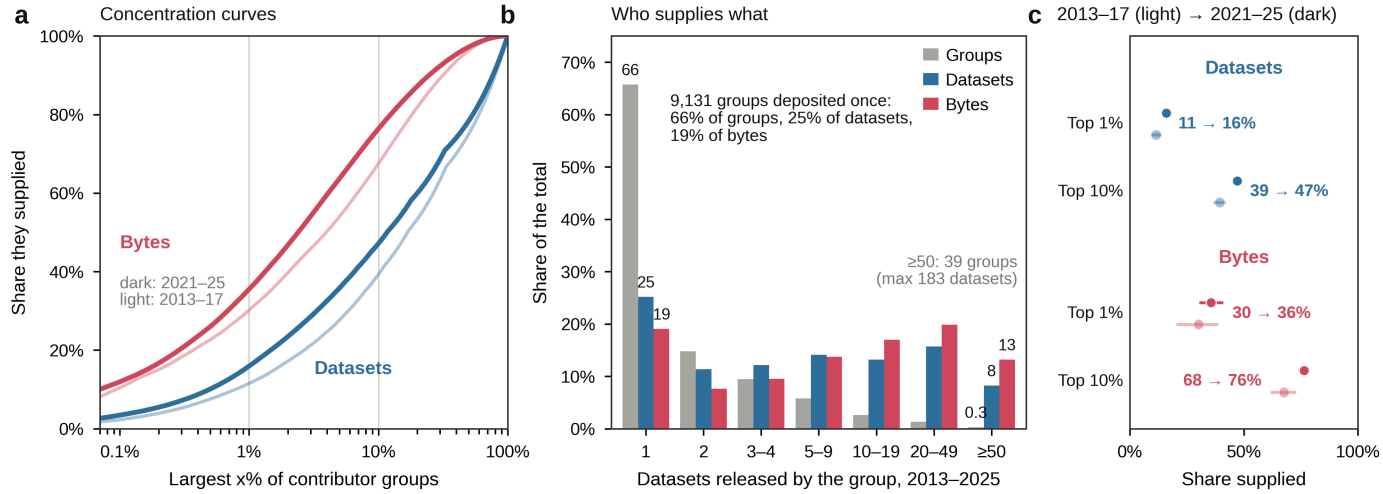
Recorded model label	Reference; qualification where needed
LTQ Orbitrap	14
LTQ	66
LTQ Orbitrap XL	67
4800 Proteomics Analyzer	68
QSTAR	69
LTQ Orbitrap Velos	70
TripleTOF 5600	71
LTQ Orbitrap Elite	72
LTQ FT	73
Q Exactive	15
Synapt MS	74 Synapt platform; the recorded label does not specify a variant.
6520A Quadrupole Time-of-Flight LC/MS	75 6520 family; cited catalog does not establish the A suffix.
ultraflex	76 LIFT-TOF/TOF family; generation is unspecified.
Q-Tof Ultima	77 Ultima Global manual; the recorded label is less specific.
Orbitrap Fusion ETD	78
maXis	79
Orbitrap Fusion	78
Q TRAP	80
TripleTOF 6600	81
impact II	82
Q Exactive Plus	83
Q Exactive HF	16

Recorded model label	Reference; qualification where needed
Orbitrap Fusion Lumos	84
Q Exactive HF-X	85
autoflex	86 Manufacturer documentation for the autoflex family.
impact HD	87
timsTOF Pro	17
SYNAPT G2-Si	88
Orbitrap Exploris 480	89
Orbitrap Eclipse	90
timsTOF Pro 2	91
Orbitrap Exploris 240	92
timsTOF	93 Original timsTOF; the deposited label may also be generic.
timsTOF SCP	91
Orbitrap Astral	94
timsTOF HT	91
Orbitrap Ascend	90

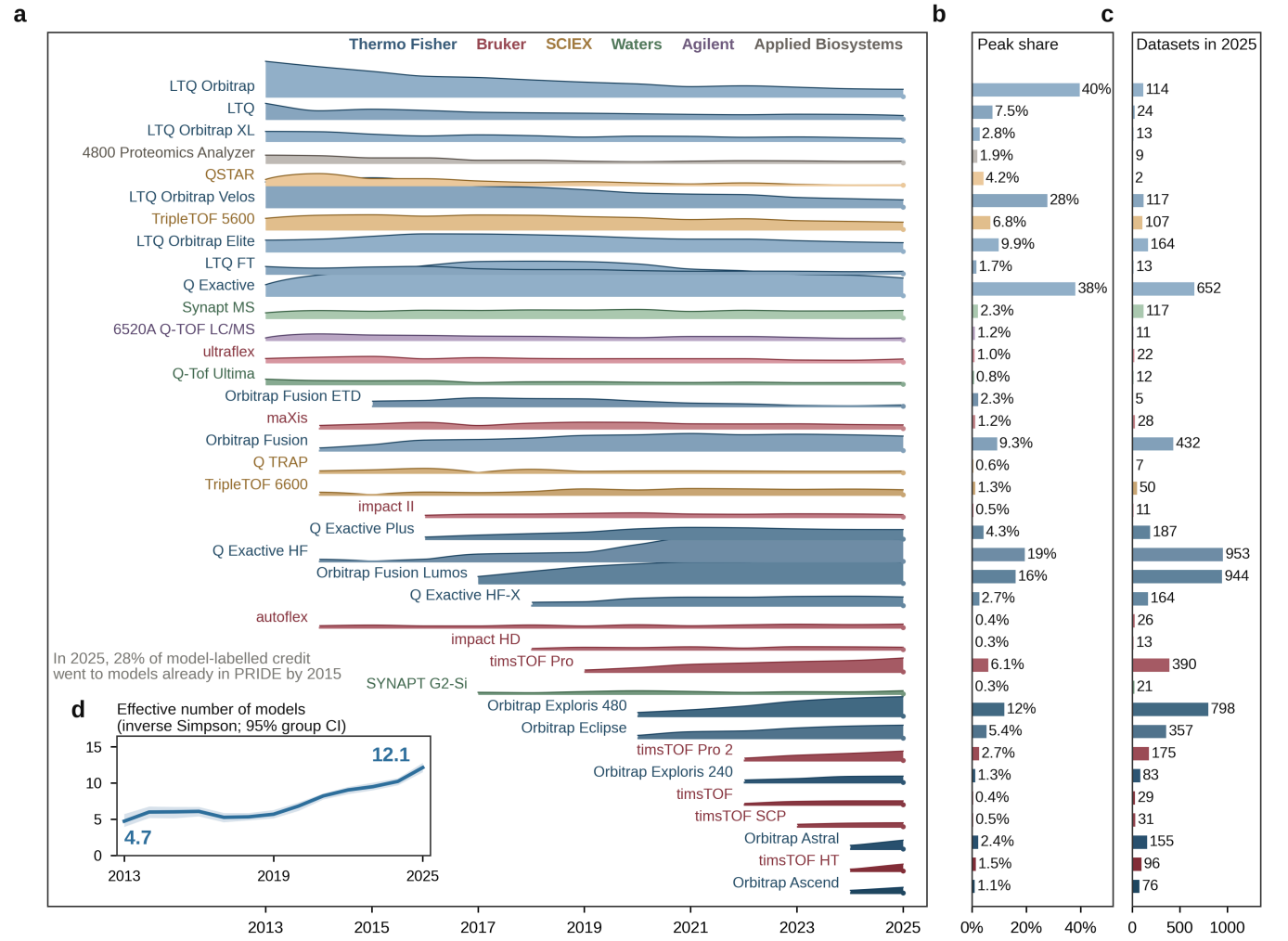
Labelling, acquisition and file formats

Method or format	Reference
TMT; TMTpro; TMTpro-18plex	13,11,95
iTRAQ	12
DIA/SWATH	27
mzIdentML; mzTab; mzML	22,23,96
SDRF sample-to-file metadata	29

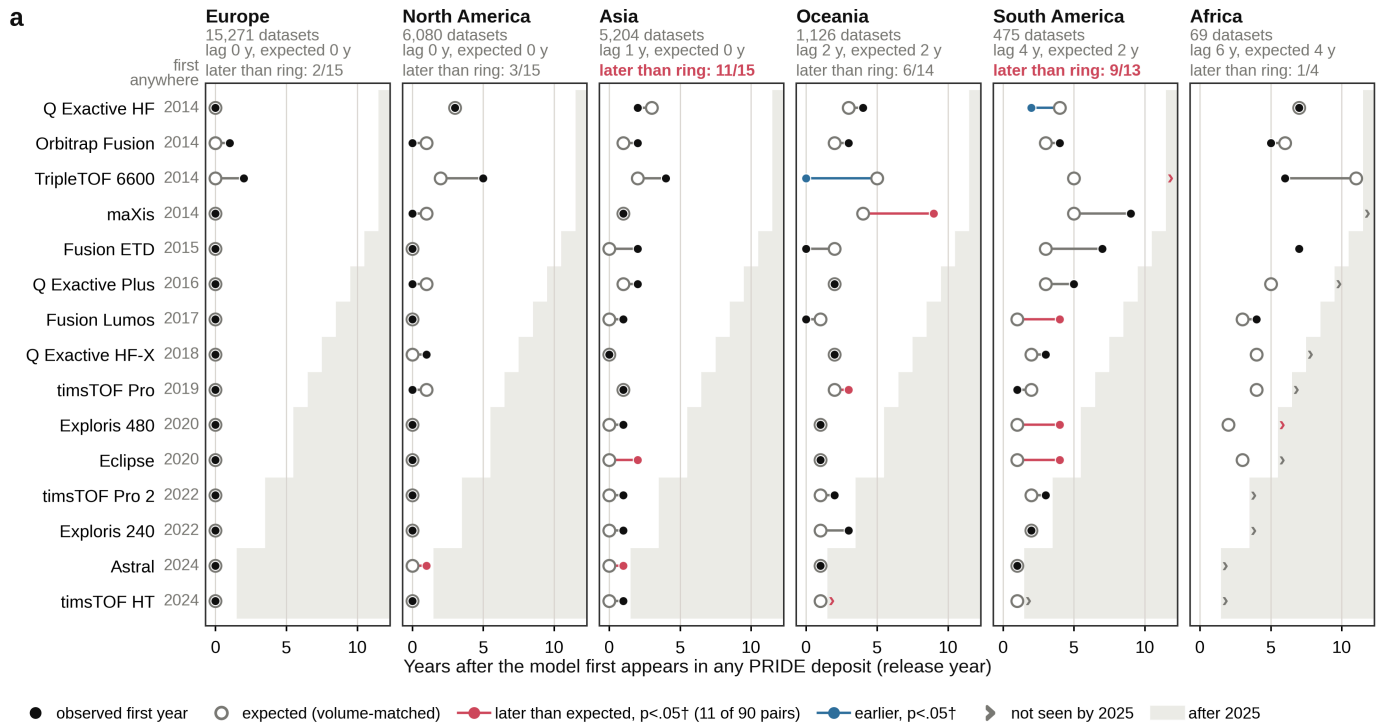
Supplementary figures



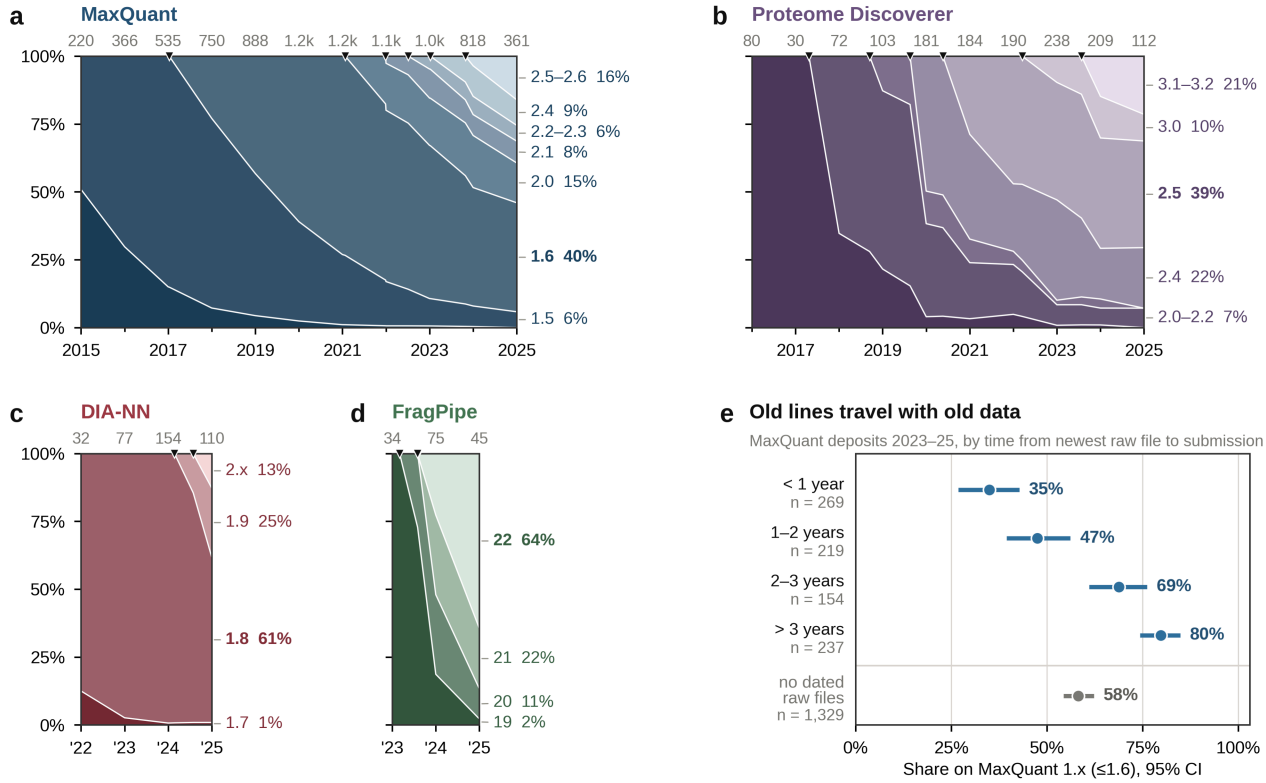
Supplementary Fig. 1 | Datasets and storage concentrated among contributor groups. a–c, The top 1% of contributor groups released 18% of PRIDE datasets and 38% of its bytes. 36,244 datasets released 2013–2025 from 13,892 contributor groups (hashed principal-investigator e-mail/ORCID; top 1% = 139 groups; 5 datasets without one principal investigator). Bytes = all files listed per dataset. Gini index of inequality in dataset counts: 0.42 in 2013–17 and 0.49 in 2021–25. Lines in c: 95% bootstrap over groups.



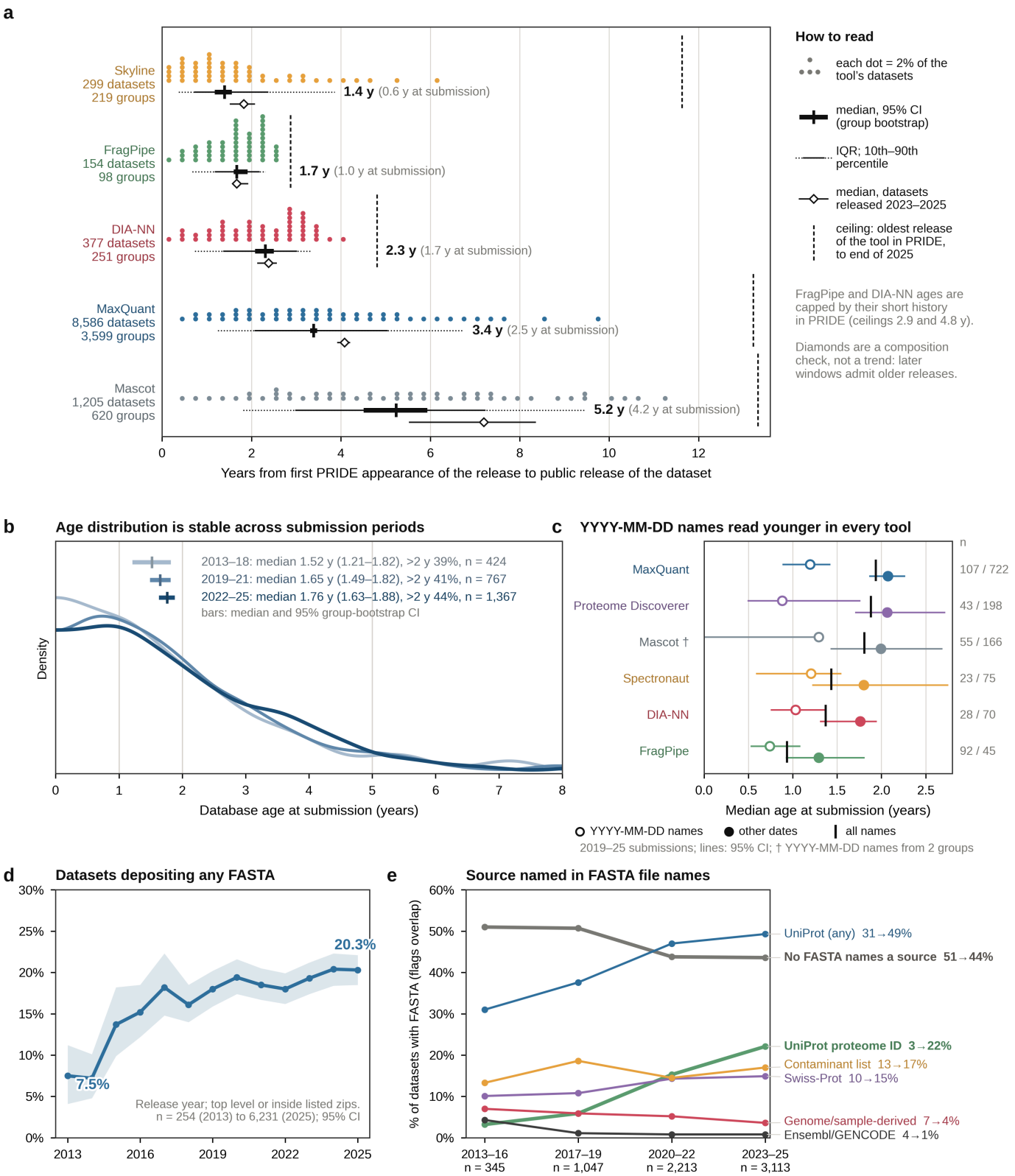
Supplementary Fig. 2 | Deposit history of common instrument models. a–d, All 37 common models still appear in 2025 deposits. Rows: models named in ≥50 public PRIDE datasets released 2013–2025 (35,770 datasets with model labels; each dataset contributes equally to its named models), ordered by first year with ≥3 datasets. Ridge height = √(share of that year's model-labelled datasets), common scale. Thin line = years with ≥1 dataset. For rows starting in 2013, earlier appearances fall outside the study period. An instrument label does not establish ownership; its first recorded year does not establish a purchase date.



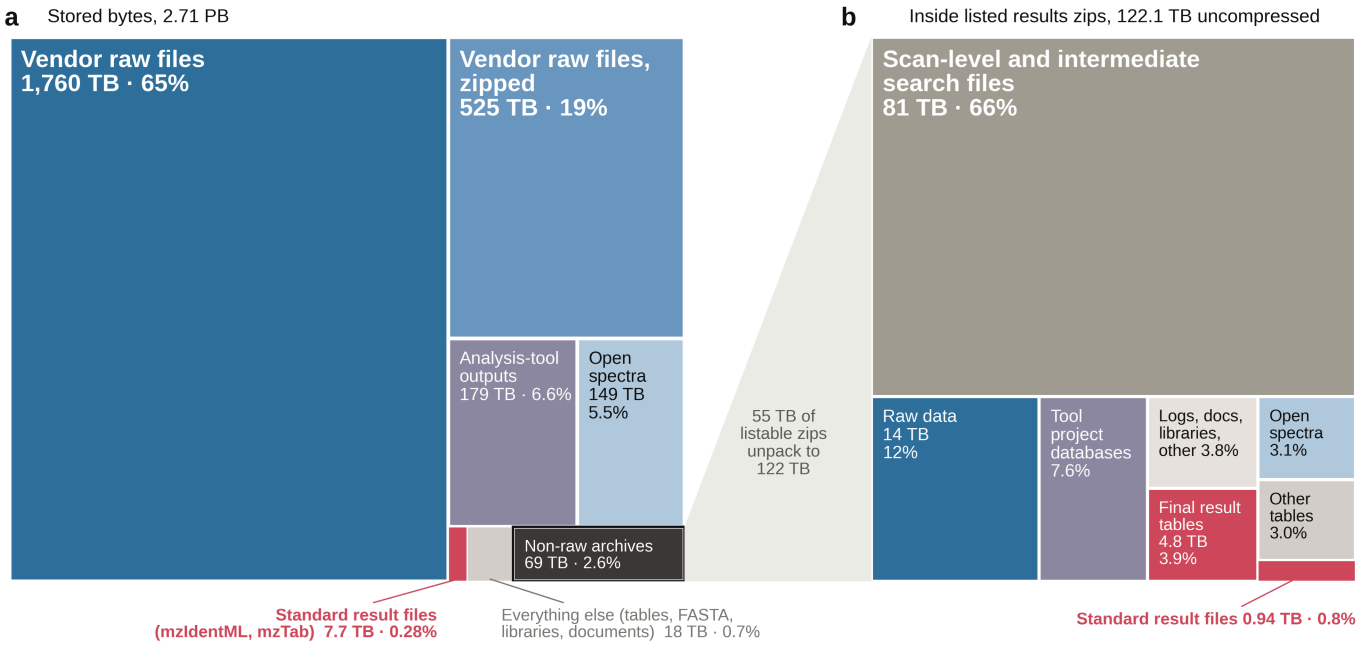
Supplementary Fig. 3 | First deposits naming new instrument models, by continent. a, Africa's 6-year delay is close to that expected from its 69 datasets. First release year of 15 models (≥100 datasets; first ≥5 in 2014 or later) in each continent, based on principal-investigator country (28,225 of 36,249 datasets). Delay is measured from each model's first deposit anywhere and summarised as the median across models observed. Ring: expected median first year if instrument labels were independent of continent within each year, given each continent's dataset count. A deposit is not a purchase record. † Hypergeometric comparison assumes independent datasets. It ignores clustering within contributor groups, making p-values too small (≈4.5 of 90 pairs expected below .05 by chance).



Supplementary Fig. 4 | Software version series in deposited files. a–e, Among 2025 MaxQuant deposits with a recoverable version, 40% used the 1.6 series, first seen eight years earlier. Share of PRIDE datasets recording each software version series, by submission year (datasets released 2013–2025). Versions are read from deposited parameter files, logs and workflows (MaxQuant, DIA-NN, FragPipe) or mzIdentML writer fields (Proteome Discoverer). Black triangles: the first record of a version series by a second contributor group. Grey numbers: datasets with a version. 2025 coverage: MaxQuant 361 of 576 datasets naming the tool, Proteome Discoverer 112 of 486, DIA-NN 110 of 331, FragPipe 45 of 142. a–d: submission (deposit) year on one time scale; 2025 = submissions public by 31 Dec 2025. e: 95% group-bootstrap intervals over hashed contributor groups.



Supplementary Fig. 5 | Ages of software releases and sequence databases. a, At public release, median software-release age is ≥5.2 years for Mascot and ≥1.4 for Skyline. PRIDE datasets released 2013–2025 whose deposited files record an exact software release (one value per dataset and tool). Age = public release date minus the first PRIDE submission recording the same release. Software exists on or before that date, so every age is a lower bound. Using the first record by a second contributor group instead gives median ages of Skyline 1.2, FragPipe 1.5, DIA-NN 2.1, MaxQuant 3.3 and Mascot 4.9 years. Excluded: 2 datasets whose PRIDE release date precedes the first-seen submission; 4 Skyline files whose "version" was "64-bit". **b–e,** Deposited sequence databases are typically 1.7 years old, and 42% are older than two years. PRIDE datasets released 2013–2025. b, c: age of the main sequence database (FASTA) at submission, from a high-confidence date in its file name (2,558 datasets, 1,305 groups; median 1.67 years, 95% confidence interval 1.58–1.80; >1 year 69%, >5 years 8%). A name date may be a download, release or tool build date, so ages are approximate.

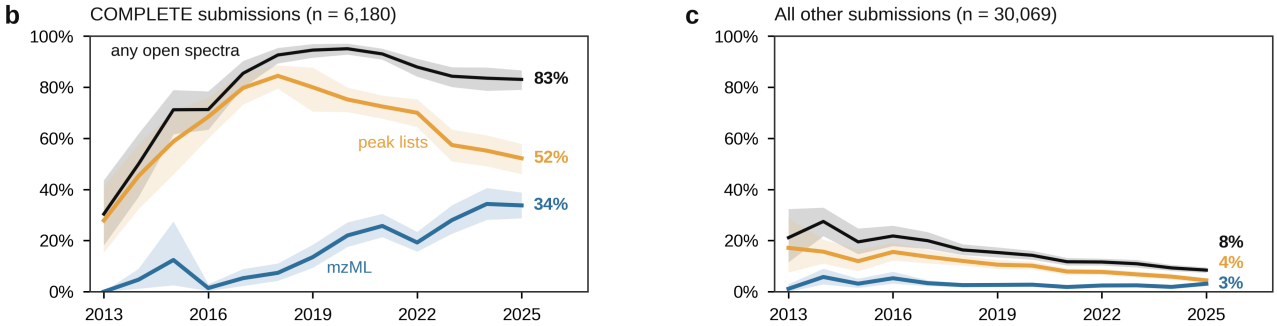


Supplementary Fig. 6 | Storage occupied by different file types. a–b, Raw data occupy 84% of PRIDE's 2.71 petabytes; final result tables occupy 4% of uncompressed results-archive storage. (a) Stored bytes of all 3.03 million top-level files in the 36,249 study datasets, by file category. (b) Uncompressed bytes inside 45,994 results zip archives with readable file lists (55.3 of 68.9 TB of non-raw archives; rar, 7z and tar were not listed), classified by file role from file names. The two panels use different scales.

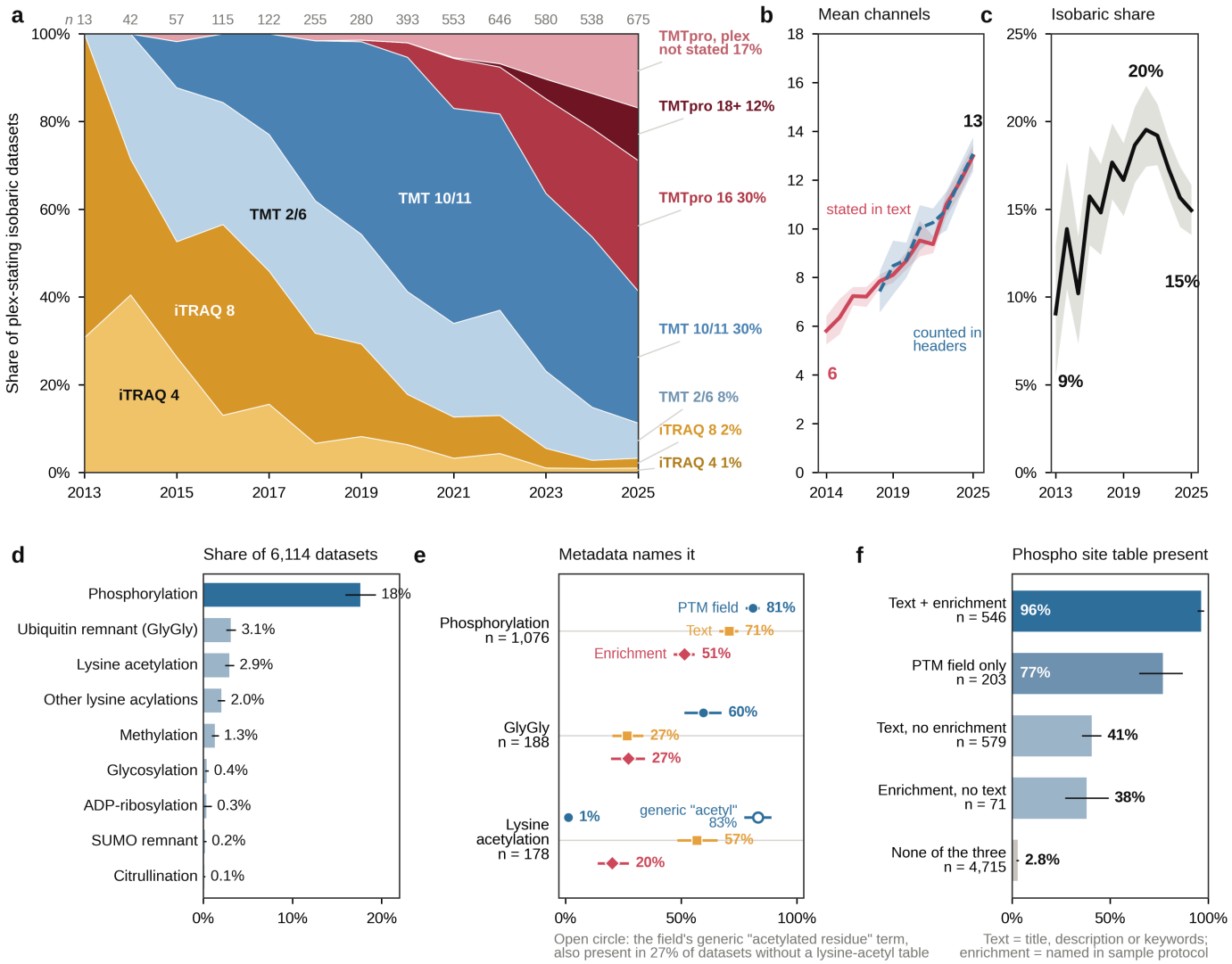
a

		2013	'14	'15	'16	'17	'18	'19	'20	'21	'22	'23	'24	2025	13–16 → 23–25 pp	
Vendor raw	**Any vendor raw**	**70**	**82**	**87**	**89**	**90**	**91**	**92**	**93**	**94**	**93**	**94**	**94**	**91**	85 → 92	+7
	Thermo .raw	67	73	78	82	84	84	85	86	88	85	85	84	78	78 → 82	+4
	Bruker .d	1.2	2.5	2.0	1.3	1.4	2.0	2.1	2.7	3.4	3.7	5.5	7.4	11	2 → 8	+7
	SCIEX .wiff	3.9	8.5	7.9	6.9	5.3	6.7	6.1	4.7	4.1	4.7	3.2	3.3	2.7	7 → 3	-4
Open spectra	**Any open spectra**	**24**	**35**	**36**	**33**	**31**	**30**	**27**	**27**	**25**	**25**	**23**	**20**	**20**	34 → 21	-13
	mzML	0.8	5.4	6.0	4.4	3.7	3.4	4.3	5.7	5.8	5.5	6.6	6.7	7.7	5 → 7	+2
	mzXML	2.8	6.2	5.3	3.3	5.0	3.1	4.2	2.8	3.0	2.1	2.9	2.2	2.2	4 → 2	-2
	Peak lists (MGF…)	20	26	27	28	25	25	21	20	19	19	15	13	12	27 → 13	-14
Results	**Standard results**	9.4	13	19	17	17	18	17	19	19	20	18	17	16	16 → 17	+1
	Tool-native outputs	31	31	28	32	31	35	36	41	43	42	44	45	44	31 → 44	+14
Context	FASTA	6.3	6.4	12	13	16	15	17	18	17	17	18	18	18	11 → 18	+7
	SDRF	0.4	2.3	2.8	1.5	2.0	0.6	0.9	0.8	0.5	1.0	2.0	3.0	3.5	2 → 3	+1
	README	0.8	1.7	0.3	0.6	1.0	0.5	0.7	1.1	0.7	0.6	0.9	0.7	0.9	1 → 1	+0

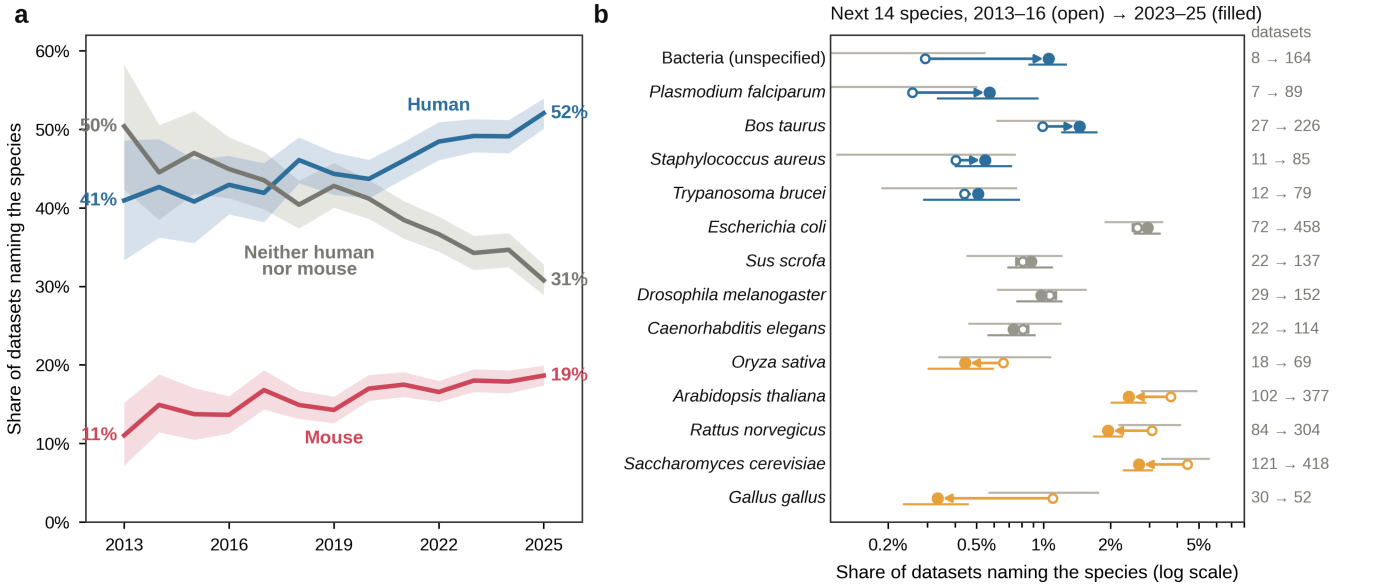
% of datasets with the file type, by release year; right: period shares and change (percentage points)



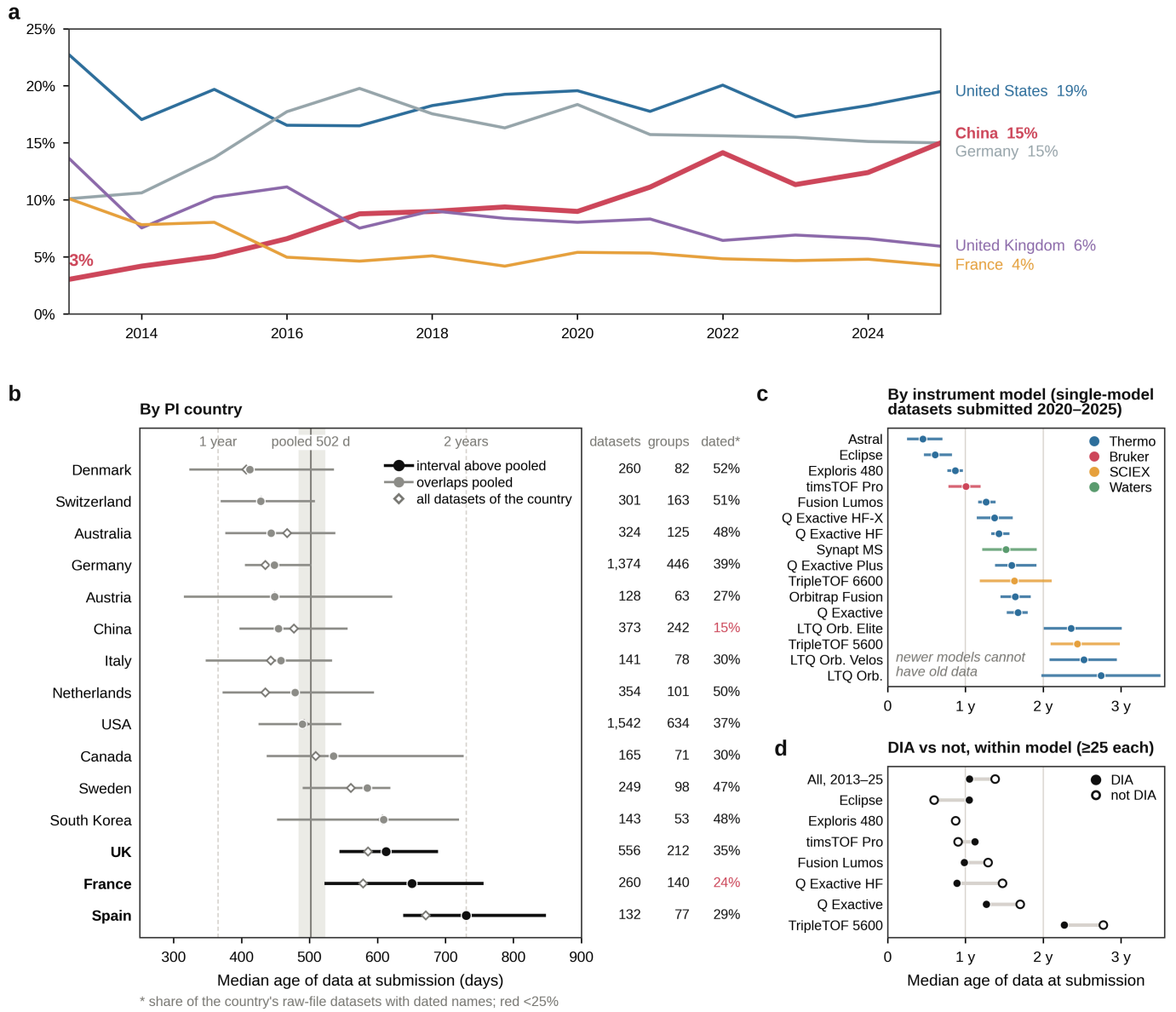
Supplementary Fig. 7 | File formats. a–c, Datasets with open spectrum formats fell from 34% to 21% as COMPLETE submissions halved; the share with mzML increased. PRIDE datasets 2013–2025 (n = 36,249) with at least one top-level file of each type (file names; compression stripped; /generated/ paths and zip contents excluded). Periods: 2013–16 (n = 2,726) vs 2023–25 (n = 15,565). Standard results are submitter-deposited mzIdentML or mzTab files; repository-generated files are excluded. b–c, open spectra by PRIDE submission type (COMPLETE submissions must include peak lists). COMPLETE fell from 29% to 15% of datasets, accounting for 5–10 of the 13-point fall when changes in submission-type composition and within-type rates were calculated in both orders. Shading: 95% CI, group bootstrap. File presence does not establish validity or reuse.



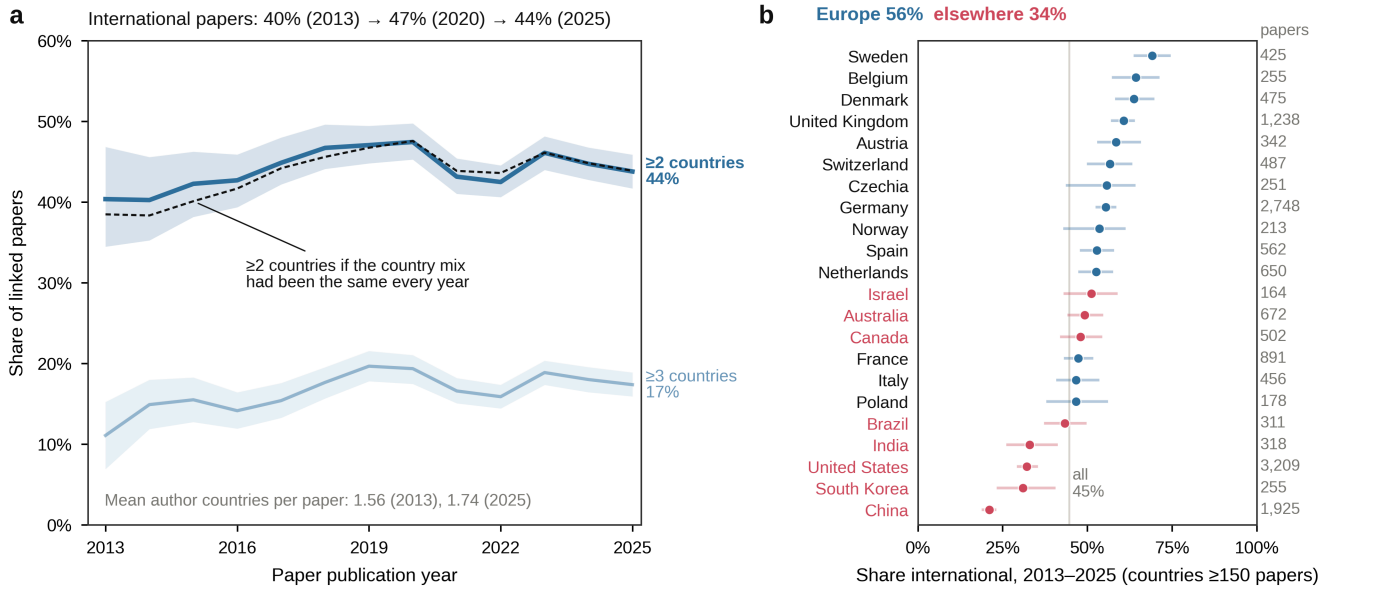
Supplementary Fig. 8 | Isobaric labelling and protein modifications. a–c, iTRAQ gave way to TMTpro, and the mean number of sample channels per labelled dataset grew from 6 to 13. PRIDE datasets with isobaric labelling in metadata, 2013–2025 (n = 6,116). a, Number of sample channels (plex) stated in text, among the 4,269 datasets that state one (highest TMT plex stated is used); iTRAQ 61% → 4% and TMTpro 0% → 48% from 2013–16 to 2023–25. b, Mean channels per dataset: counts stated in text vs reporter channels counted independently in table headers (channels used ≤ kit size; years with ≥20 datasets). c, isobaric datasets among all datasets. Shading: 95% CI, group bootstrap over contributor groups. **d–f,** PRIDE's post-translational modification (PTM) field names the modification in 81% of datasets with phosphorylation site tables and 60% with diglycine (GlyGly) site tables. PRIDE datasets 2013–2025 whose deposited MaxQuant output shows site tables (n = 6,114; the default Oxidation (M)Sites table makes them visible). A site table shows that the search allowed the modification as a variable modification; it does not show that samples were enriched for it. d, modifications with a site table. e, of datasets with that site table, share whose metadata names the modification. f, Share with a phosphorylation site table, by which metadata sources name phosphorylation. Whiskers: 95% CI, group bootstrap over contributor groups. Absence means "not detected". MaxQuant only.



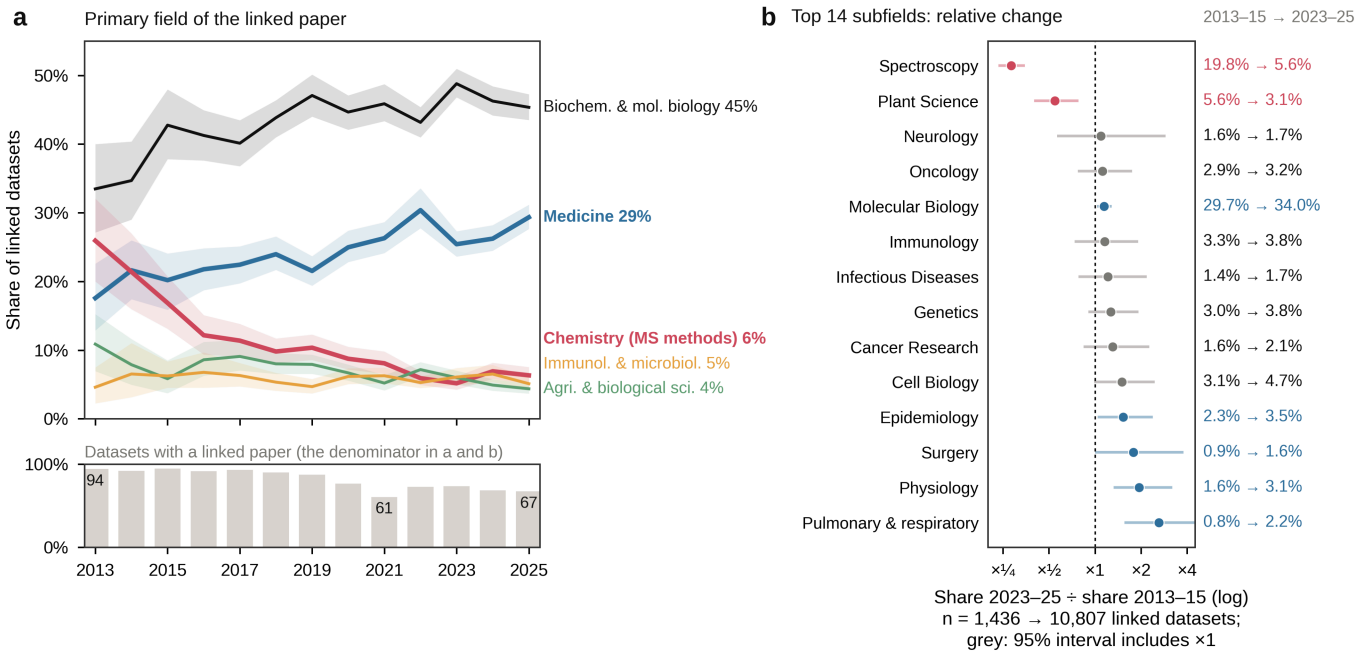
Supplementary Fig. 9 | Organisms. a–b, PRIDE deposits became more concentrated on human and mouse: datasets naming neither fell from 46% to 33%. PRIDE datasets 2013–2025 (n = 36,249; 8 list no species). Species from PRIDE records, with strains grouped under their species name; a dataset naming several species (2,931) counts once for each, so shares do not sum to 100%. Title and b: 2013–16 (n = 2,726) vs 2023–25 (n = 15,565). Within the 354 contributor groups depositing in both periods, the share naming neither human nor mouse also fell, 40% (34–46) → 32% (27–38). Shading and bars: 95% CI, group bootstrap. Dataset counts rose for every species shown; only their shares fell.



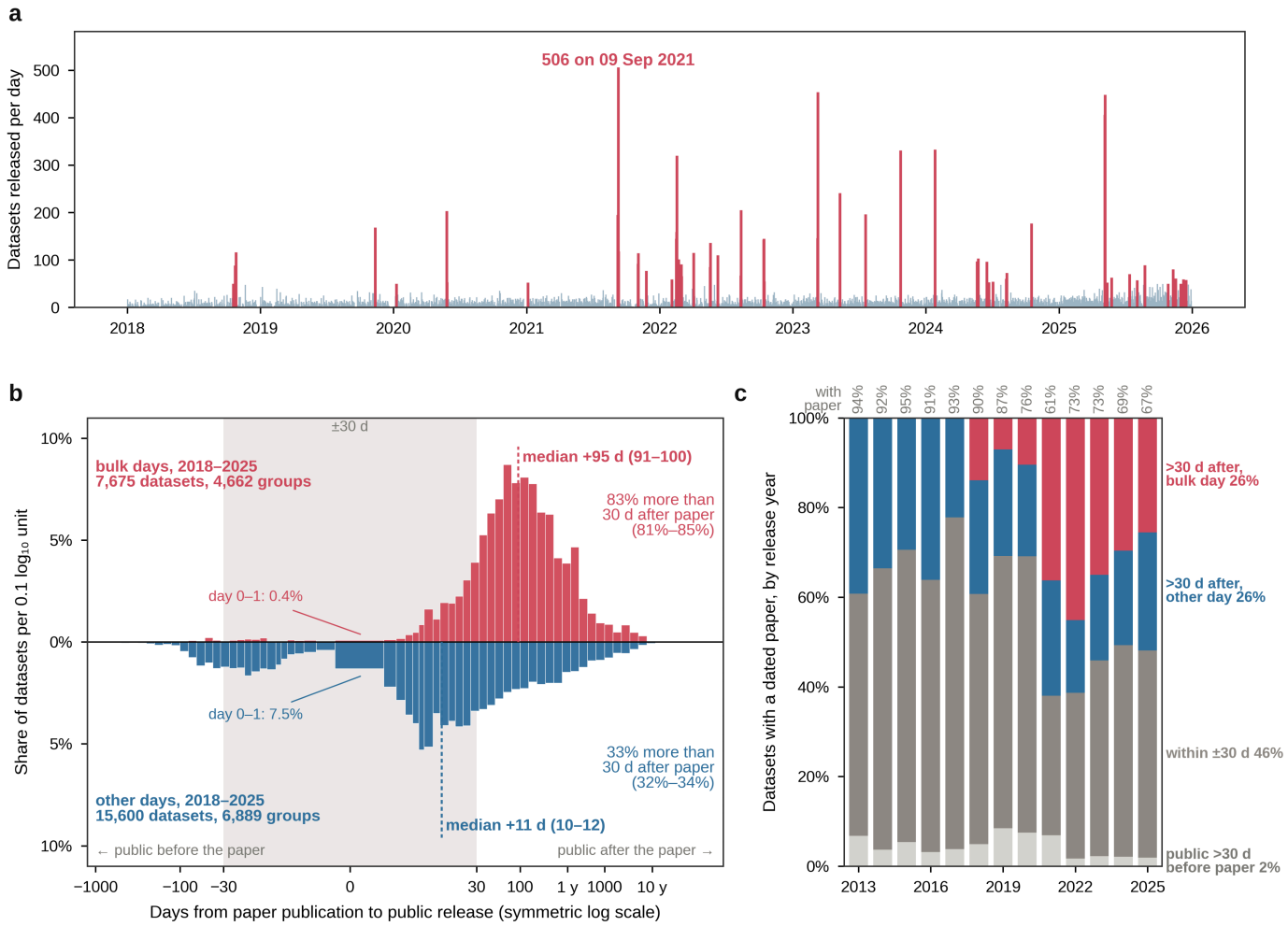
Supplementary Fig. 10 | Countries. a, China's share of public PRIDE datasets grew fivefold. Share of PRIDE datasets with an assigned country, by contributor country and release year (28,225 datasets). **b–d,** In the selected comparison, median data age at submission is 1.7–2.0 years for the UK, France and Spain, against 1.4 overall. Days from the latest acquisition date in raw-file names to PRIDE submission; only 11,868 of 35,364 datasets with raw files (34%) have dated names. (b) Thermo Fisher only, no detected data-independent acquisition (DIA), submitted 2016–2025, countries with ≥100 such datasets (9,109; the pooled median covers all of them). No country's median is clearly below the pooled median. Dots: median; bars: 95% interval over 2,000 resamples of contributor groups. A file-name date is supplied by the laboratory and may record naming rather than acquisition.



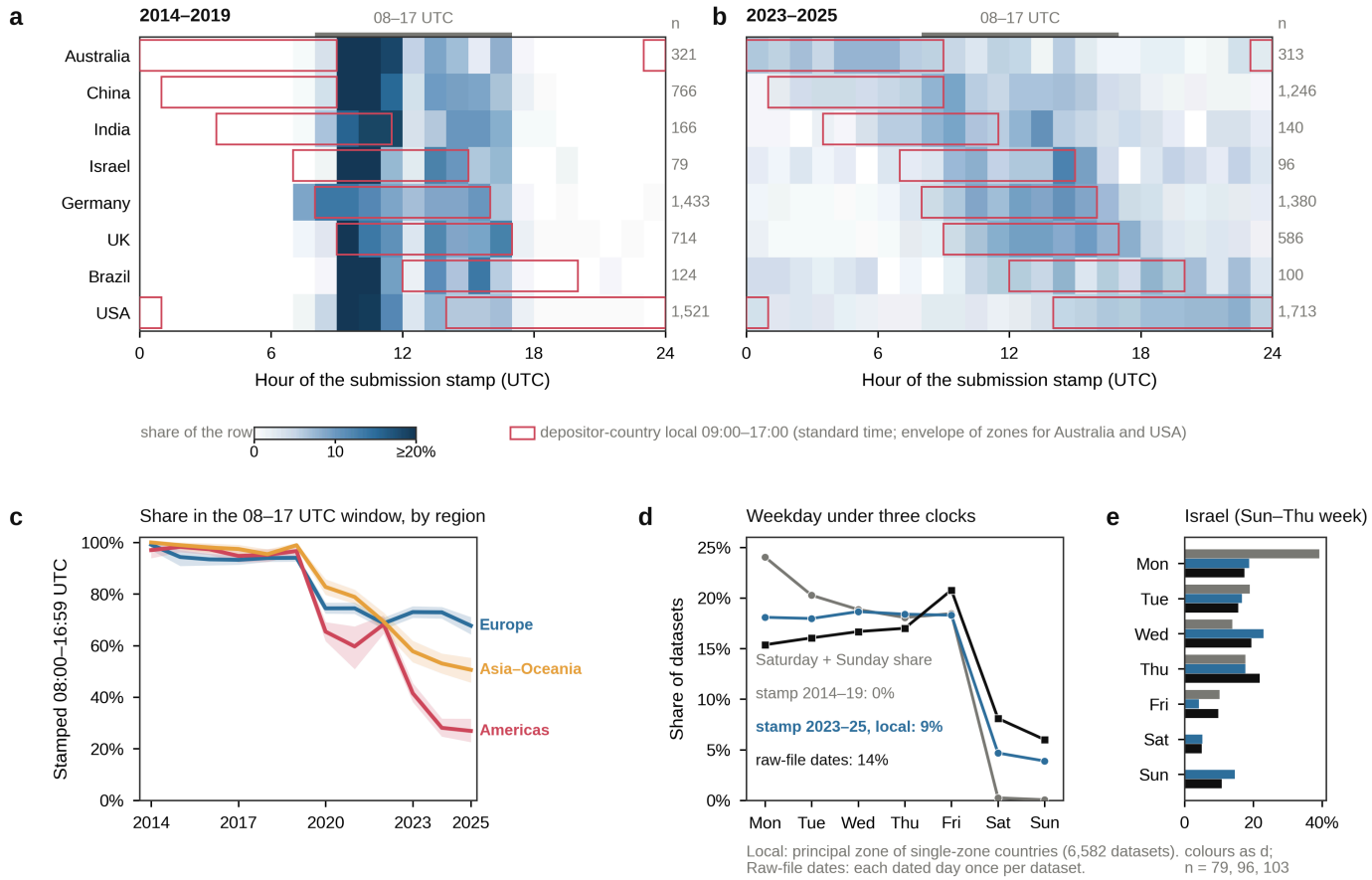
Supplementary Fig. 11 | Collaboration. a–b, International coauthorship occurs in 56% of European groups' papers, 32% of US groups' papers and 21% of Chinese groups' papers. Each paper is counted once: 22,939 OpenAlex works published 2013–2025 and linked as the primary paper of a PRIDE dataset (of 23,000; 61 lack author countries). International = authors' institutions in ≥2 countries. Country in b = the depositing group's country. Bands and bars: 95% bootstrap over groups.



Supplementary Fig. 12 | Research fields of linked papers. a–b, The share of datasets linked to mass-spectrometry methods papers fell from 26% to 6%; medicine rose from 18% to 29%. OpenAlex primary field/subfield of the paper linked to each PRIDE dataset, by release year: 27,031 of 36,249 datasets (2013–2025) have a linked paper. "Methods" = subfield Spectroscopy (95% of Chemistry). Each dataset is counted once; bands and bars: 95% bootstrap intervals over contributor groups.



Supplementary Fig. 13 | Release timing. a, Public releases cluster on a few dates: 25% of datasets since 2018 went public on 62 days. PRIDE datasets made public per day, 2018–2025. Red: days with ≥50 releases. Datasets released on those days went public a median 95 days after their paper (other days: 11). **b–c,** The share released >30 days after publication rose from 32% to 56% after 2020; the increase occurred on days with many releases. Public release minus publication date of the linked paper (OpenAlex): 27,001 of 36,209 datasets have a dated paper. Days with many releases: ≥50 PRIDE releases (62 days, all since Oct 2018). Release years 2013–20 vs 2021–25: >30 days after publication on other days 26% vs 22%; on days with many releases 7% vs 34%. Intervals: 95% over 2,000 resamples of contributor groups. Descriptive only: who set a release date is not recorded.



Supplementary Fig. 14 | Recorded submission times. a–e, Until 2019, 95% of PRIDE submissions were time-stamped 08–17 Coordinated Universal Time (UTC), wherever they came from. Submission timestamp of 35,946 of 36,249 datasets (the rest carry no time of day), as stored (UTC). From about 2020, recorded times follow depositors' working days (52% within 08–17 UTC in 2025). What the timestamp records is not documented; weekday patterns before 2020 describe this recorded clock rather than depositors' working patterns.